

```
In [1]: import numpy as np # linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)
import seaborn as sns
sns.set(style="whitegrid")
import matplotlib.pyplot as plt
from collections import Counter
%matplotlib inline
import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))
```

```
In [2]: # ignore warnings
import warnings
warnings.filterwarnings('ignore')
```

```
In [3]: fifa19 = pd.read_csv(r"C:\Users\suras\OneDrive\NARESH IT\Practice tasks\FIFA.csv",
```

```
In [4]: fifa19.head()
```

```
Out[4]:
```

	ID	Name	Age	Photo	Nationality	
0	158023	L. Messi	31	https://cdn.sofifa.org/players/4/19/158023.png	Argentina	https://cc
1	20801	Cristiano Ronaldo	33	https://cdn.sofifa.org/players/4/19/20801.png	Portugal	https://cc
2	190871	Neymar Jr	26	https://cdn.sofifa.org/players/4/19/190871.png	Brazil	https://cc
3	193080	De Gea	27	https://cdn.sofifa.org/players/4/19/193080.png	Spain	https://cc
4	192985	K. De Bruyne	27	https://cdn.sofifa.org/players/4/19/192985.png	Belgium	https://c

5 rows × 88 columns



```
In [5]: fifa19.info()
```

```

<class 'pandas.core.frame.DataFrame'>
Int64Index: 18207 entries, 0 to 18206
Data columns (total 88 columns):
ID                18207 non-null int64
Name              18207 non-null object
Age              18207 non-null int64
Photo            18207 non-null object
Nationality       18207 non-null object
Flag             18207 non-null object
Overall          18207 non-null int64
Potential        18207 non-null int64
Club             17966 non-null object
Club Logo        18207 non-null object
Value            18207 non-null object
Wage             18207 non-null object
Special          18207 non-null int64
Preferred Foot    18159 non-null object
International Reputation 18159 non-null float64
Weak Foot        18159 non-null float64
Skill Moves      18159 non-null float64
Work Rate        18159 non-null object
Body Type        18159 non-null object
Real Face        18159 non-null object
Position         18147 non-null object
Jersey Number    18147 non-null float64
Joined           16654 non-null object
Loaned From      1264 non-null object
Contract Valid Until 17918 non-null object
Height           18159 non-null object
Weight           18159 non-null object
LS               16122 non-null object
ST               16122 non-null object
RS               16122 non-null object
LW               16122 non-null object
LF               16122 non-null object
CF               16122 non-null object
RF               16122 non-null object
RW               16122 non-null object
LAM              16122 non-null object
CAM              16122 non-null object
RAM              16122 non-null object
LM               16122 non-null object
LCM              16122 non-null object
CM               16122 non-null object
RCM              16122 non-null object
RM               16122 non-null object
LWB              16122 non-null object
LDM              16122 non-null object
CDM              16122 non-null object
RDM              16122 non-null object
RWB              16122 non-null object
LB               16122 non-null object
LCB              16122 non-null object
CB               16122 non-null object
RCB              16122 non-null object
RB               16122 non-null object

```

```

Crossing          18159 non-null float64
Finishing         18159 non-null float64
HeadingAccuracy   18159 non-null float64
ShortPassing      18159 non-null float64
Volleys           18159 non-null float64
Dribbling         18159 non-null float64
Curve            18159 non-null float64
FKAccuracy        18159 non-null float64
LongPassing       18159 non-null float64
BallControl       18159 non-null float64
Acceleration      18159 non-null float64
SprintSpeed       18159 non-null float64
Agility           18159 non-null float64
Reactions         18159 non-null float64
Balance           18159 non-null float64
ShotPower         18159 non-null float64
Jumping           18159 non-null float64
Stamina           18159 non-null float64
Strength          18159 non-null float64
LongShots         18159 non-null float64
Aggression        18159 non-null float64
Interceptions     18159 non-null float64
Positioning       18159 non-null float64
Vision            18159 non-null float64
Penalties         18159 non-null float64
Composure         18159 non-null float64
Marking           18159 non-null float64
StandingTackle    18159 non-null float64
SlidingTackle     18159 non-null float64
GKDividing        18159 non-null float64
GKHandling        18159 non-null float64
GKKicking         18159 non-null float64
GKPositioning     18159 non-null float64
GKReflexes        18159 non-null float64
Release Clause    16643 non-null object
dtypes: float64(38), int64(5), object(45)
memory usage: 12.4+ MB

```

```
In [6]: fifa19['Body Type'].value_counts()
```

```

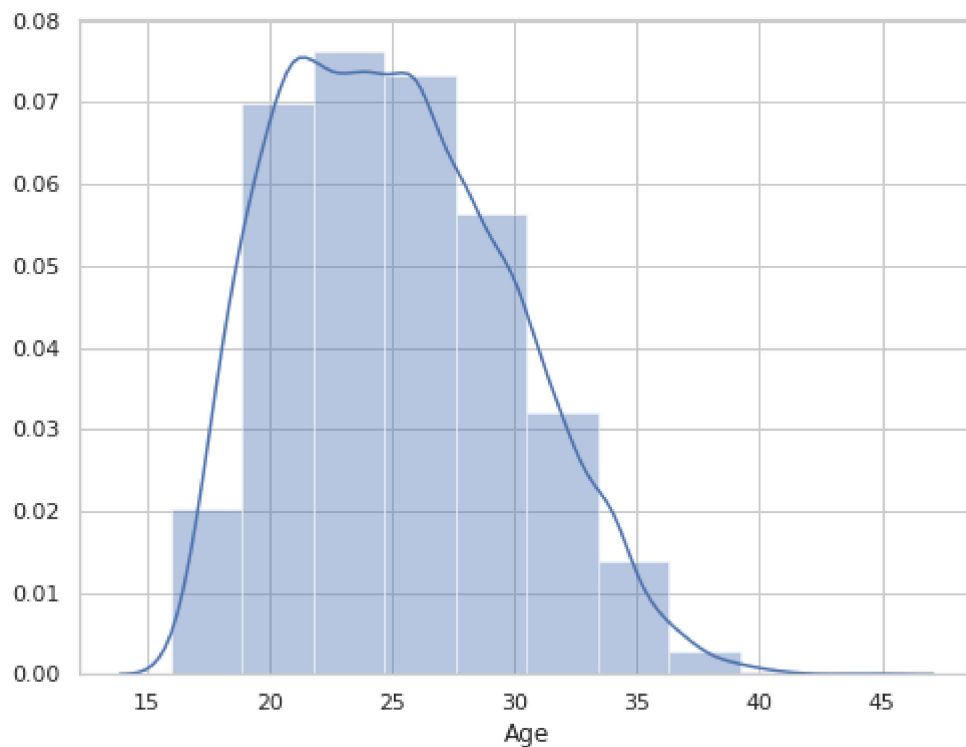
Out[6]: Normal          10595
Lean                   6417
Stocky                 1140
Akinfenwa              1
Courtois               1
Messi                  1
PLAYER_BODY_TYPE_25    1
Shaqiri                1
C. Ronaldo            1
Neymar                 1
Name: Body Type, dtype: int64

```

```

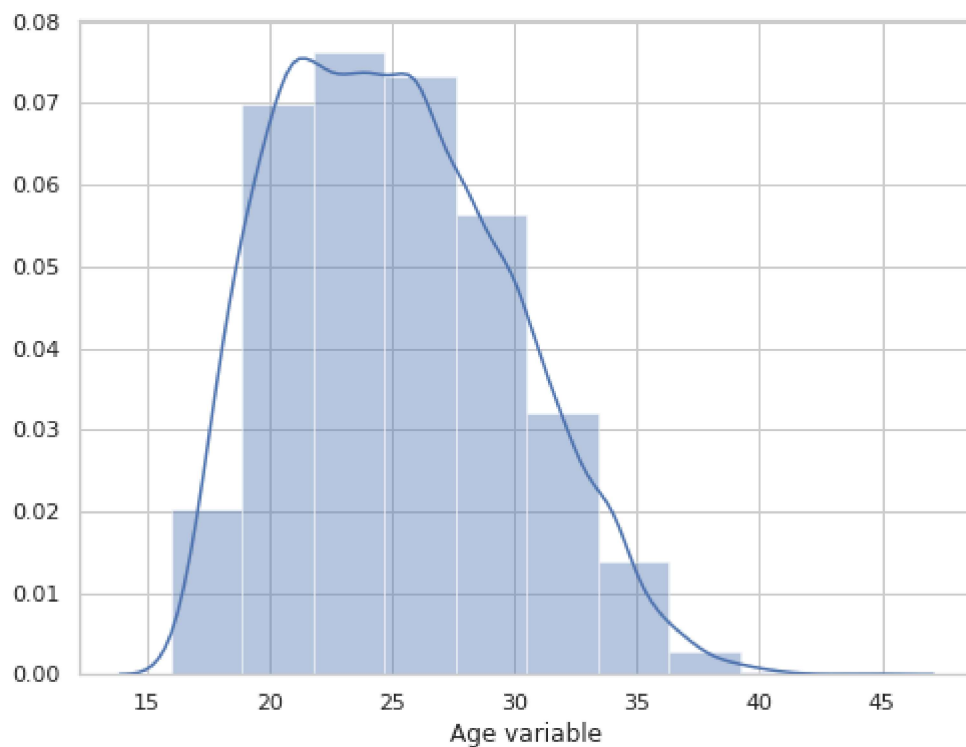
In [7]: f, ax = plt.subplots(figsize=(8,6))
x = fifa19['Age']
ax = sns.distplot(x, bins=10)
plt.show()

```



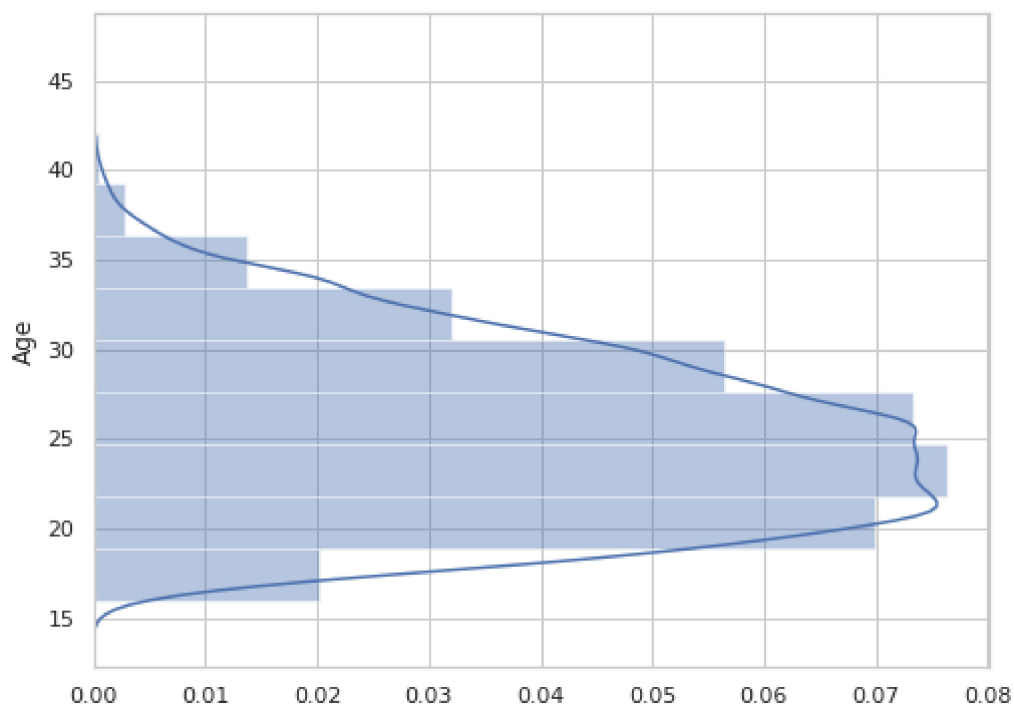
We can use Pandas series object to get an informative axis label as follows-

```
In [8]: f, ax = plt.subplots(figsize=(8,6))
x = fifa19['Age']
x = pd.Series(x, name="Age variable")
ax = sns.distplot(x, bins=10)
plt.show()
```

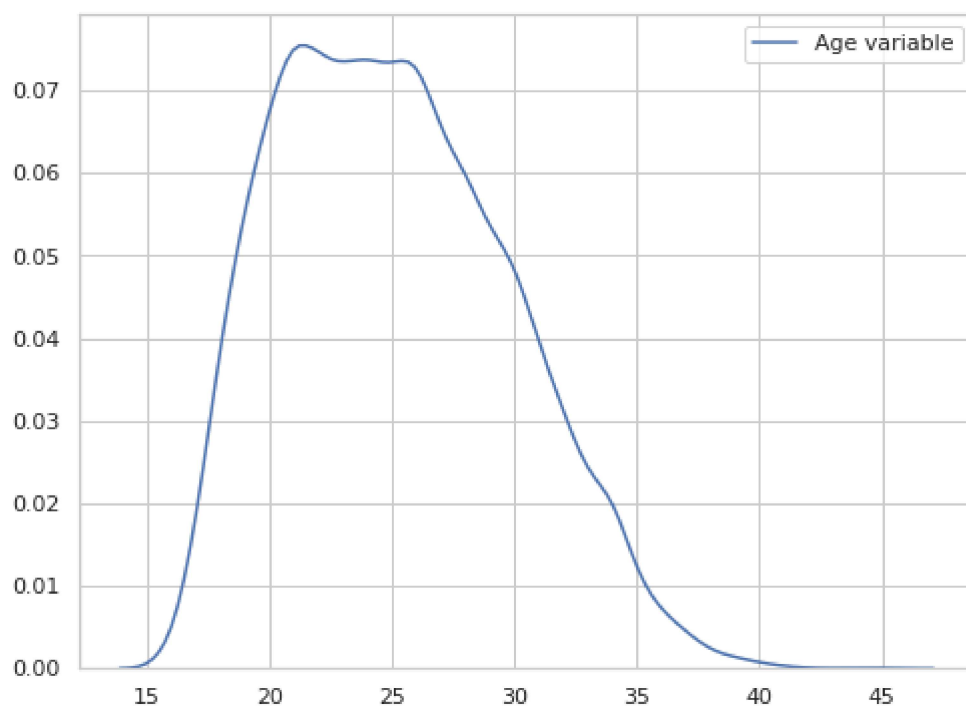


We can plot the distribution on the vertical axis as follows:-

```
In [9]: f, ax = plt.subplots(figsize=(8,6))
x = fifa19['Age']
ax = sns.distplot(x, bins=10, vertical = True)
plt.show()
```

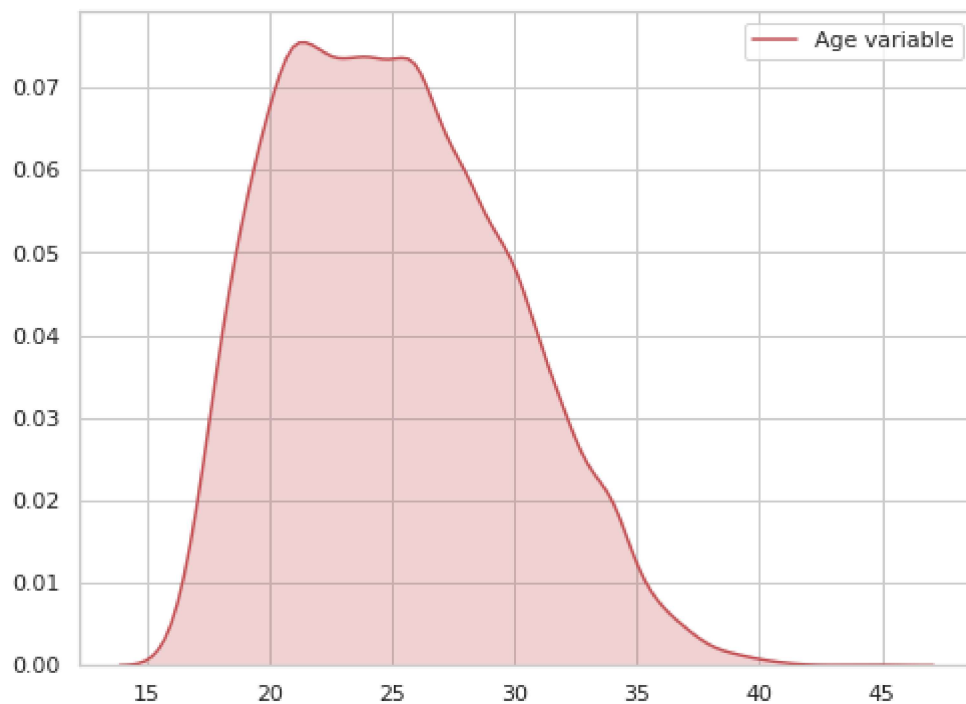


```
In [10]: f, ax = plt.subplots(figsize=(8,6))
x = fifa19['Age']
x = pd.Series(x, name="Age variable")
ax = sns.kdeplot(x)
plt.show()
```

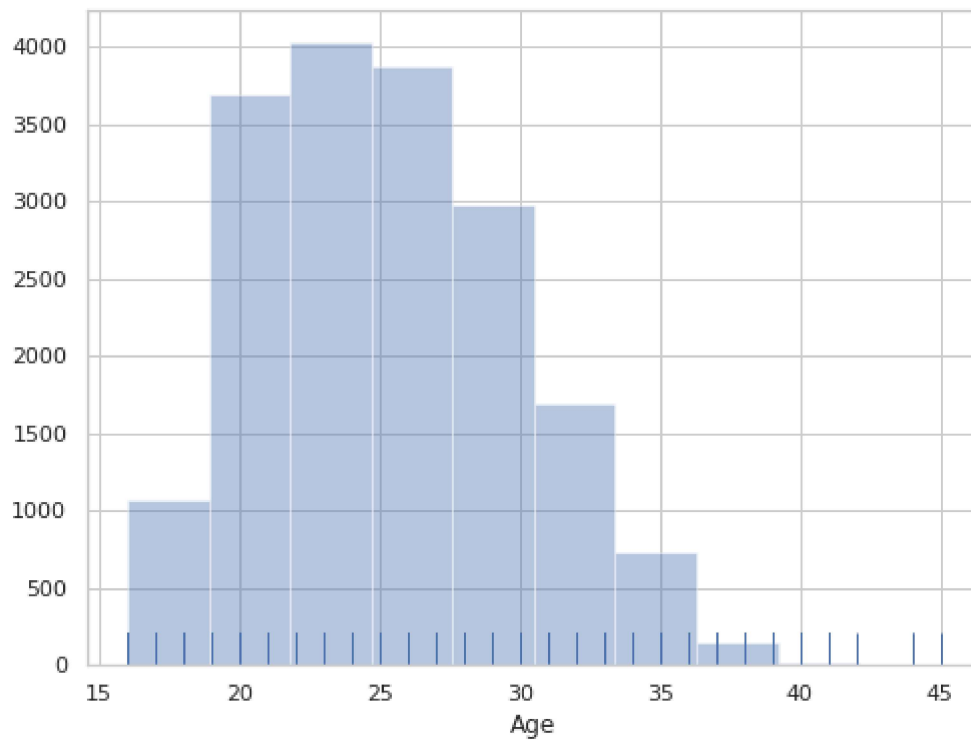


We can shade under the density curve and use a different color as follows:-

```
In [11]: f, ax = plt.subplots(figsize=(8,6))
x = fifa19['Age']
x = pd.Series(x, name="Age variable")
ax = sns.kdeplot(x, shade=True, color='r')
plt.show()
```

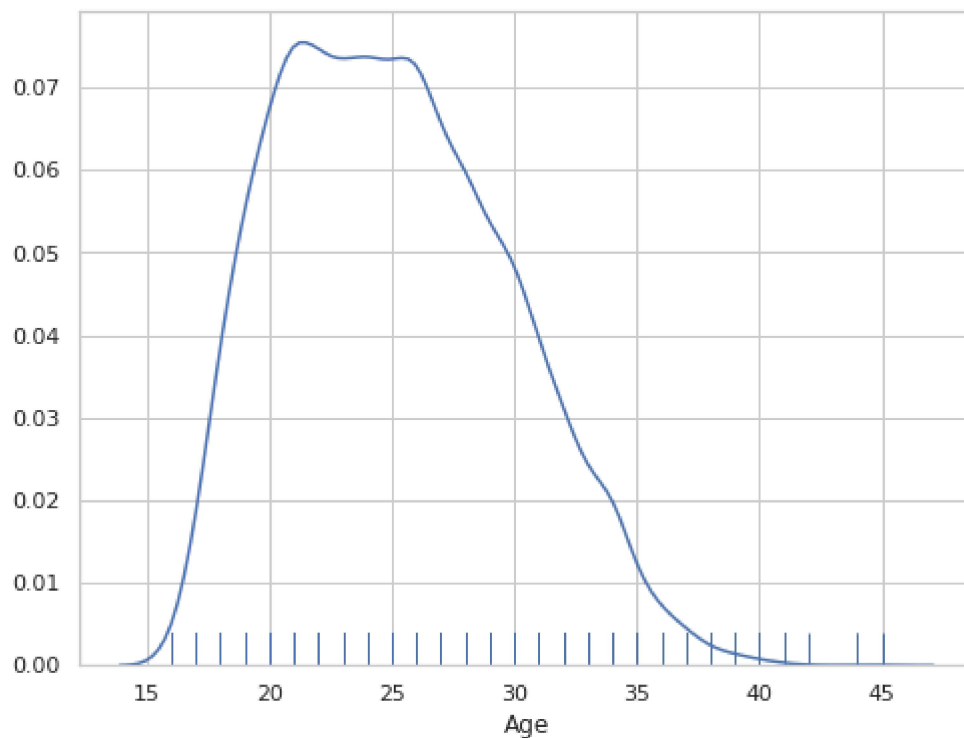


```
In [12]: f, ax = plt.subplots(figsize=(8,6))
x = fifa19['Age']
ax = sns.distplot(x, kde=False, rug=True, bins=10)
plt.show()
```



We can plot a KDE plot alternatively as follows:-

```
In [13]: f, ax = plt.subplots(figsize=(8,6))
x = fifa19['Age']
ax = sns.distplot(x, hist=False, rug=True, bins=10)
plt.show()
```



```
In [14]: fifa19['Preferred Foot'].nunique()
```

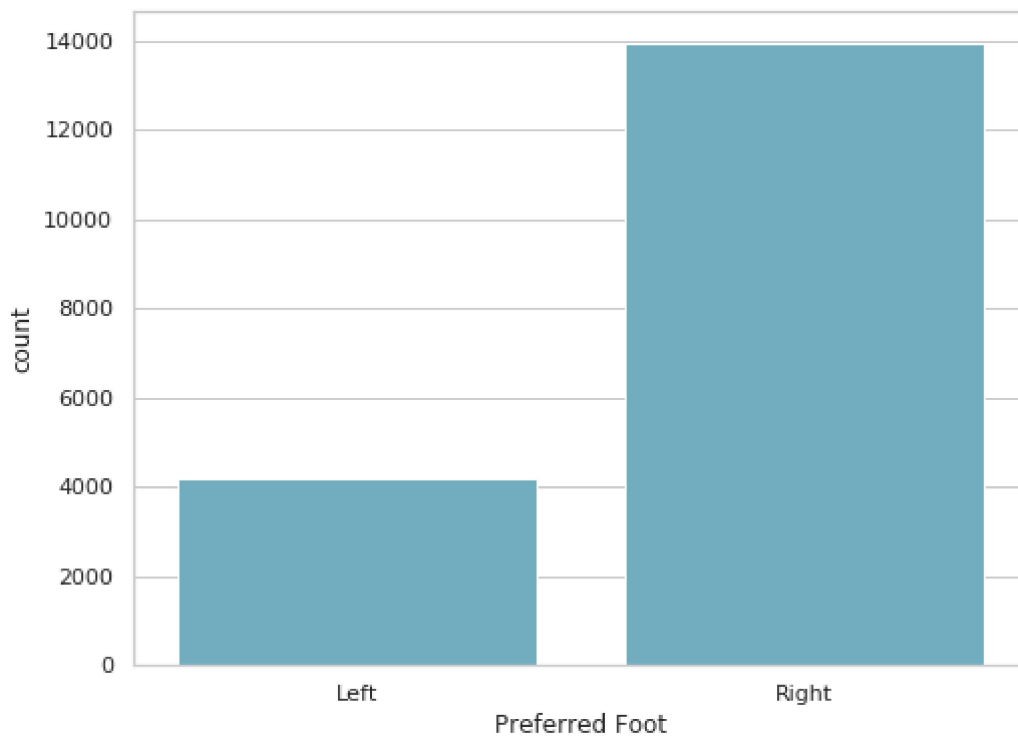
Out[14]: 2

```
In [15]: fifa19['Preferred Foot'].value_counts()
```

```
Out[15]: Right    13948  
Left       4211  
Name: Preferred Foot, dtype: int64
```

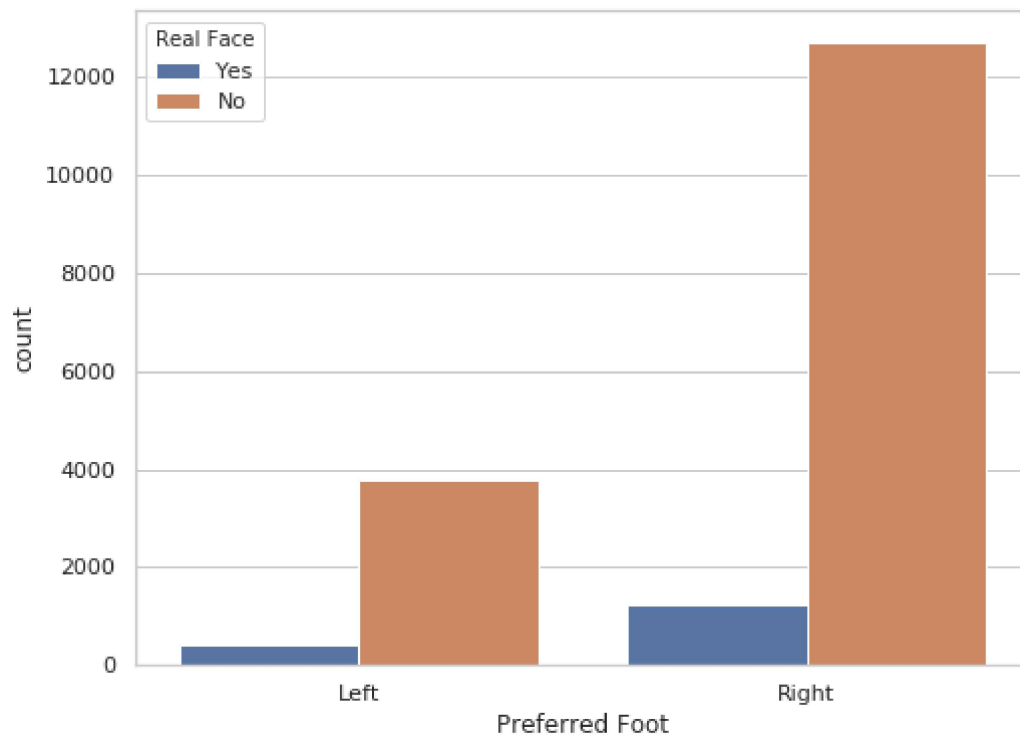
The Preferred Foot variable contains two types of values - Right and Left .

```
In [16]: f, ax = plt.subplots(figsize=(8, 6))  
sns.countplot(x="Preferred Foot", data=fifa19, color="c")  
plt.show()
```



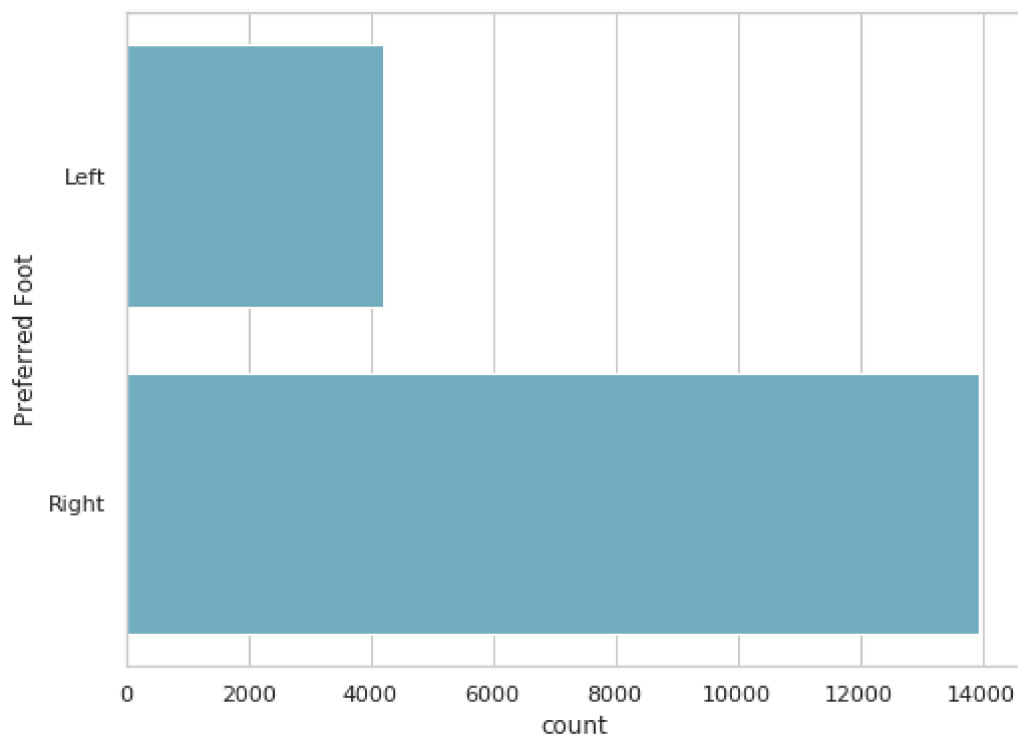
We can show value counts for two categorical variables as follows-

```
In [17]: f, ax = plt.subplots(figsize=(8, 6))  
sns.countplot(x="Preferred Foot", hue="Real Face", data=fifa19)  
plt.show()
```

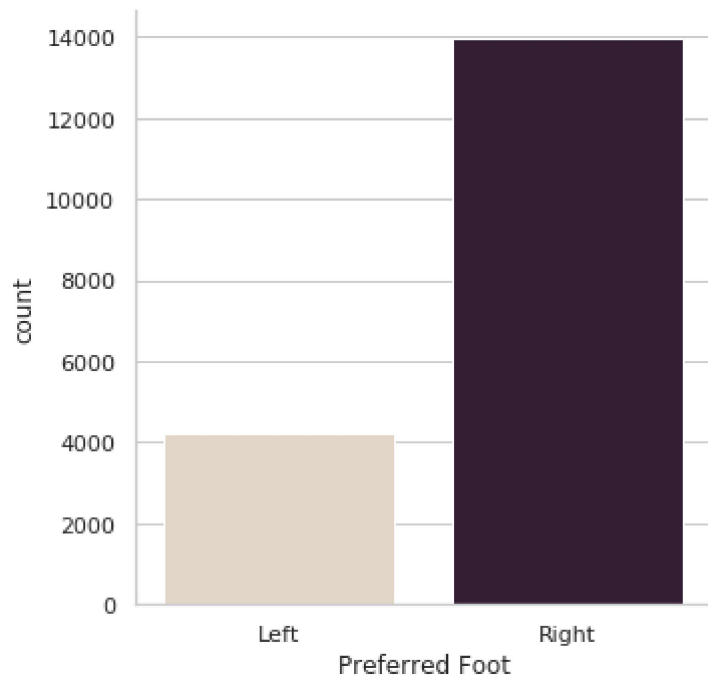


We can draw plot vertically as follows-

```
In [18]: f, ax = plt.subplots(figsize=(8, 6))  
sns.countplot(y="Preferred Foot", data=fifa19, color="c")  
plt.show()
```



```
In [19]: g = sns.catplot(x="Preferred Foot", kind="count", palette="ch:.25", data=fifa19)
```



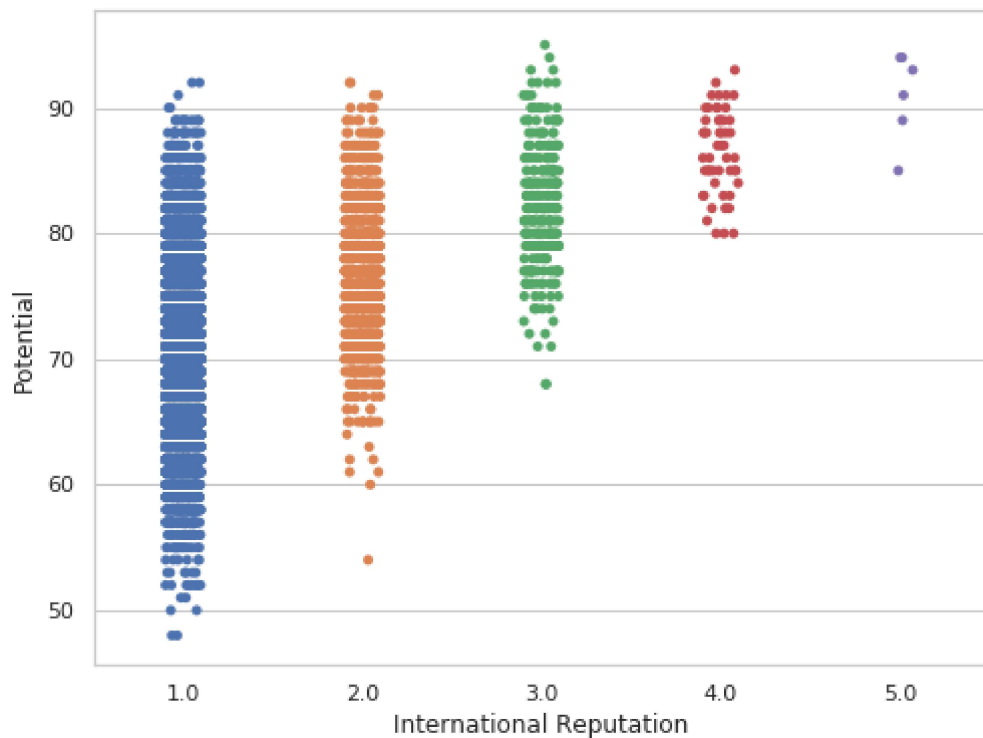
```
In [20]: fifa19['International Reputation'].nunique()
```

```
Out[20]: 5
```

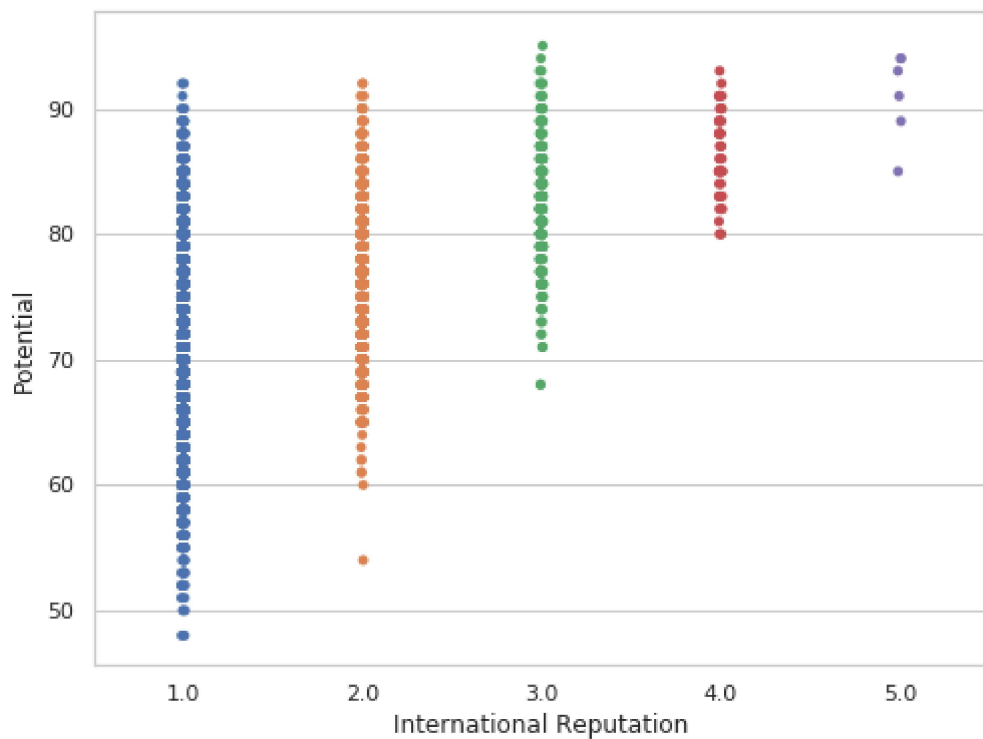
```
In [21]: fifa19['International Reputation'].value_counts()
```

```
Out[21]: 1.0    16532
         2.0     1261
         3.0      309
         4.0       51
         5.0        6
         Name: International Reputation, dtype: int64
```

```
In [22]: f, ax = plt.subplots(figsize=(8, 6))
         sns.stripplot(x="International Reputation", y="Potential", data=fifa19)
         plt.show()
```



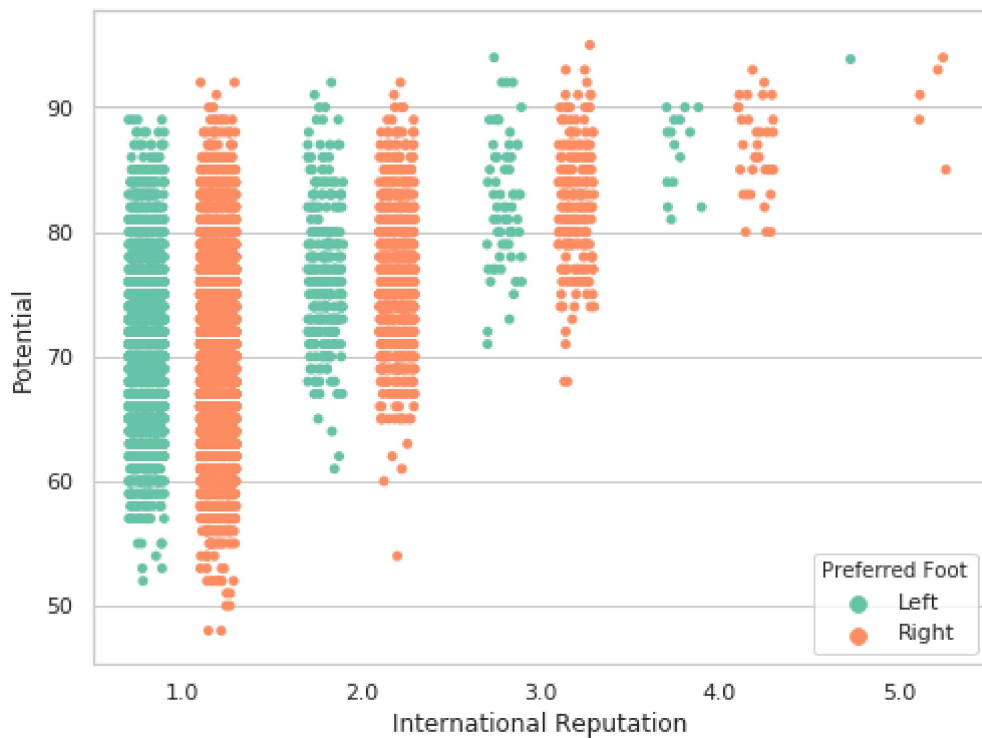
```
In [23]: f, ax = plt.subplots(figsize=(8, 6))
sns.stripplot(x="International Reputation", y="Potential", data=fifa19, jitter=0.01)
plt.show()
```



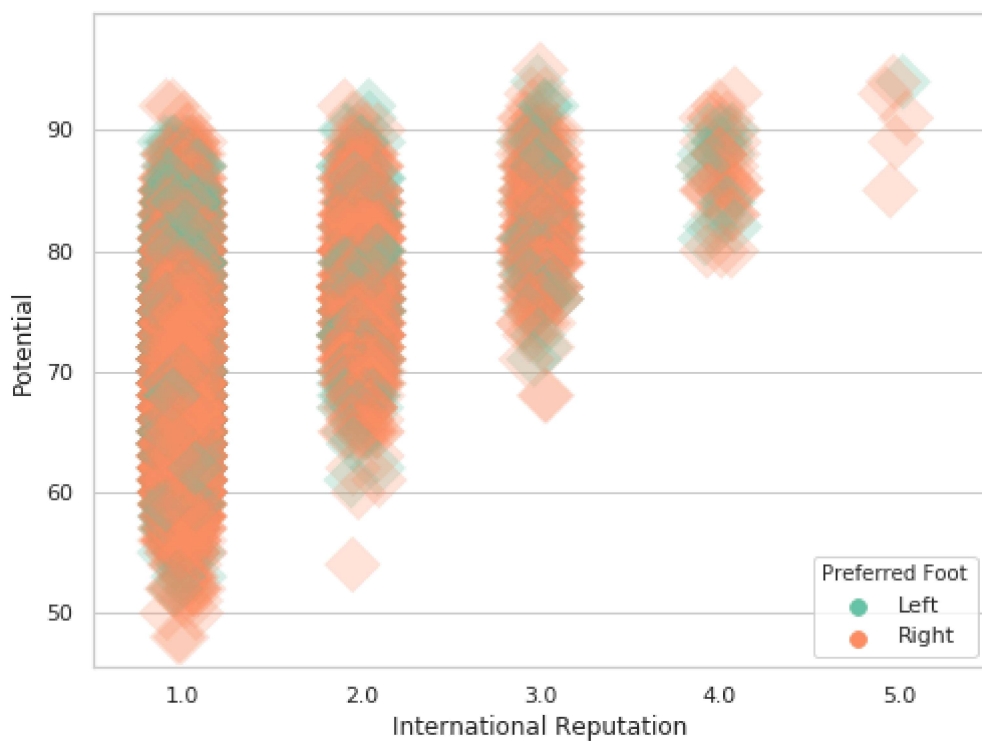
We can nest the strips within a second categorical variable - Preferred Foot - as follows-

```
In [24]: f, ax = plt.subplots(figsize=(8, 6))
sns.stripplot(x="International Reputation", y="Potential", hue="Preferred Foot",
```

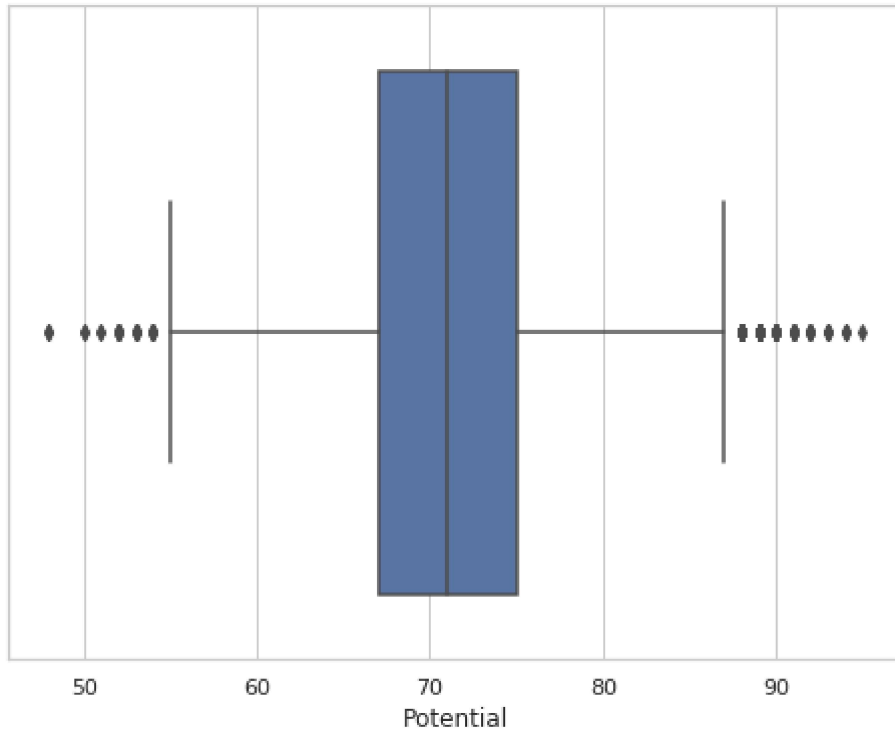
```
data=fifa19, jitter=0.2, palette="Set2", dodge=True)
plt.show()
```



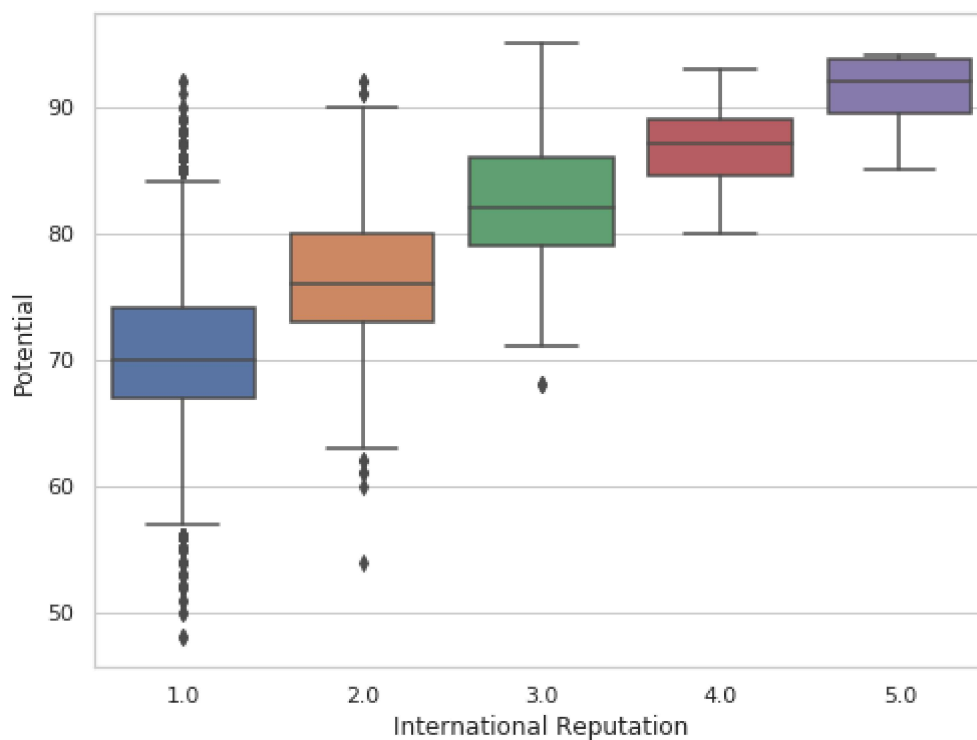
```
In [25]: f, ax = plt.subplots(figsize=(8, 6))
sns.stripplot(x="International Reputation", y="Potential", hue="Preferred Foot",
              data=fifa19, palette="Set2", size=20, marker="D",
              edgecolor="gray", alpha=.25)
plt.show()
```



```
In [26]: f, ax = plt.subplots(figsize=(8, 6))
sns.boxplot(x=fifa19["Potential"])
plt.show()
```

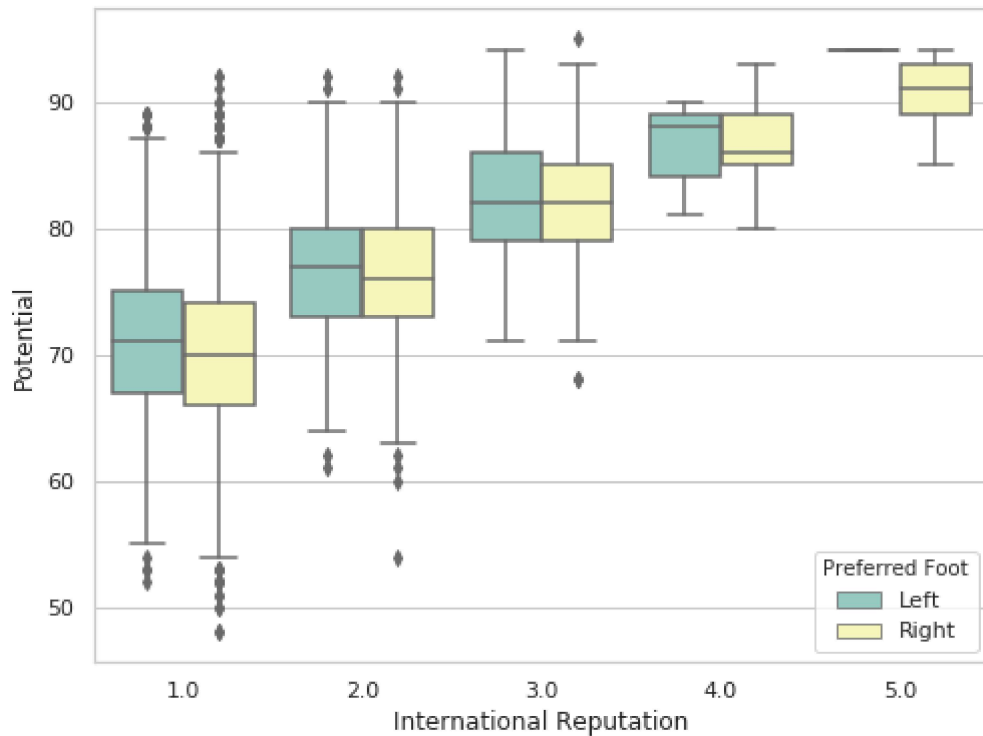


```
In [27]: f, ax = plt.subplots(figsize=(8, 6))
sns.boxplot(x="International Reputation", y="Potential", data=fifa19)
plt.show()
```

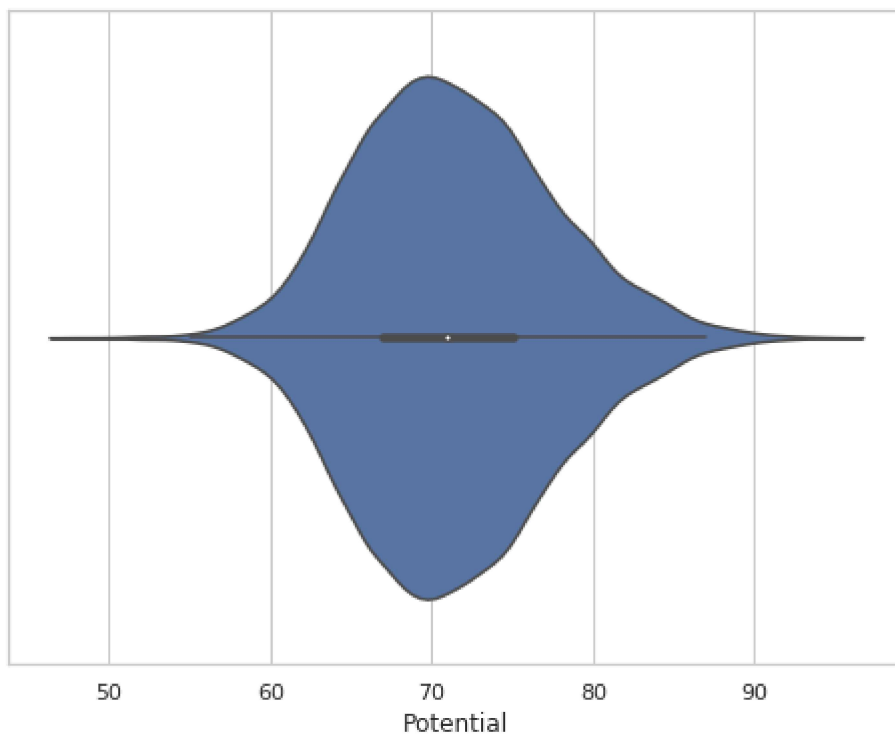


We can draw a boxplot with nested grouping by two categorical variables as follows-

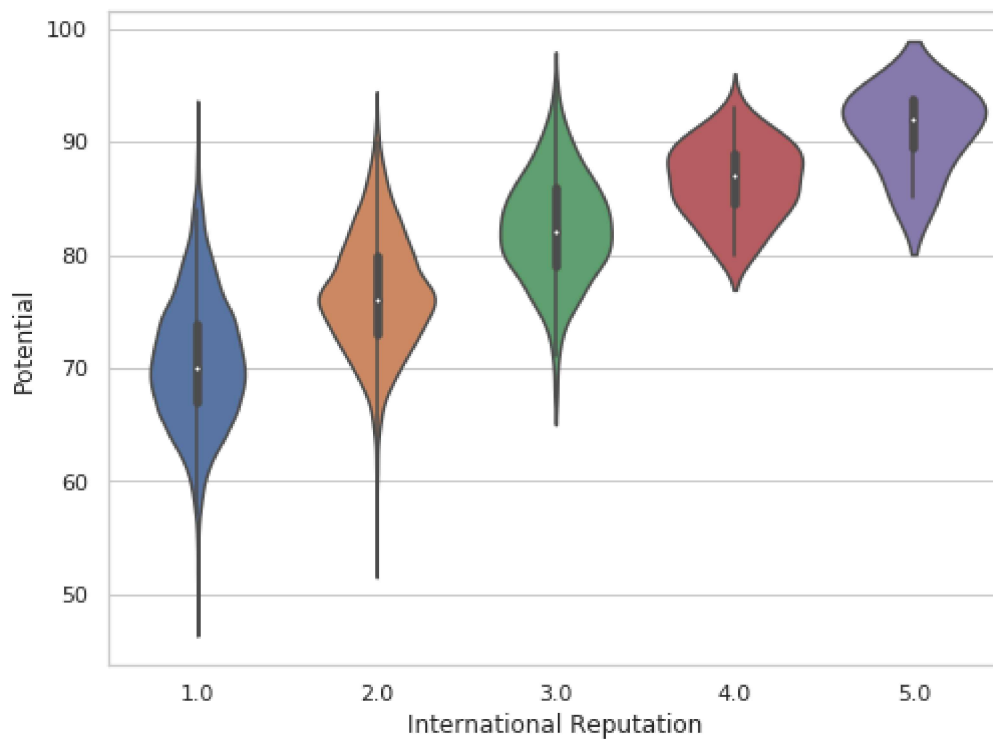
```
In [28]: f, ax = plt.subplots(figsize=(8, 6))  
sns.boxplot(x="International Reputation", y="Potential", hue="Preferred Foot", data=  
plt.show())
```



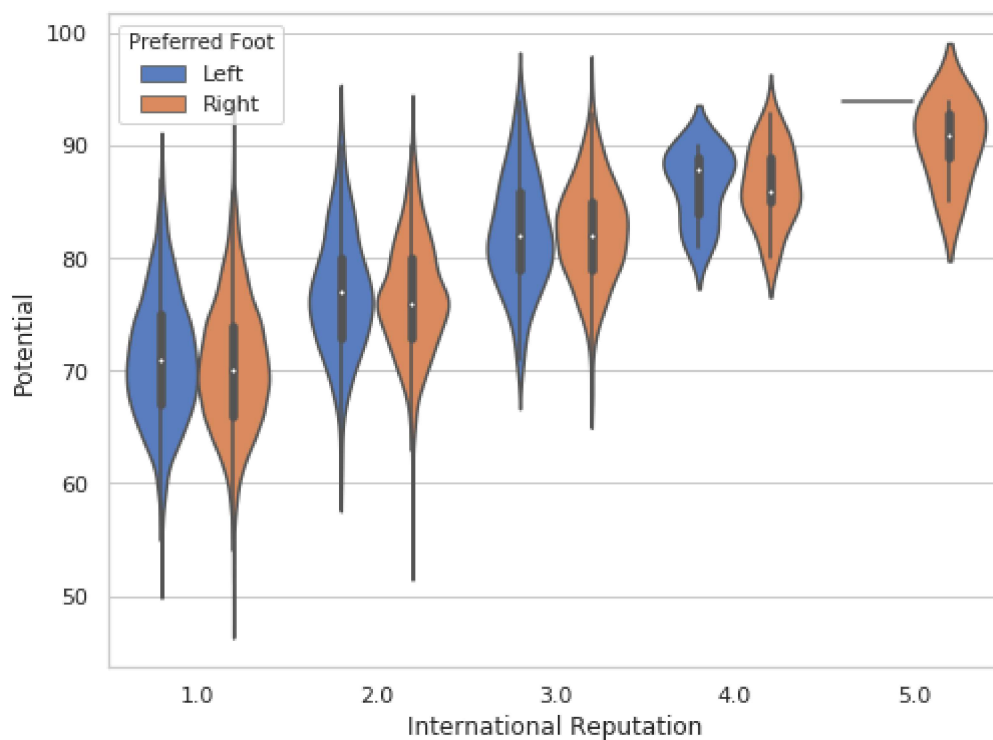
```
In [29]: f, ax = plt.subplots(figsize=(8, 6))  
sns.violinplot(x=fifa19["Potential"])  
plt.show()
```



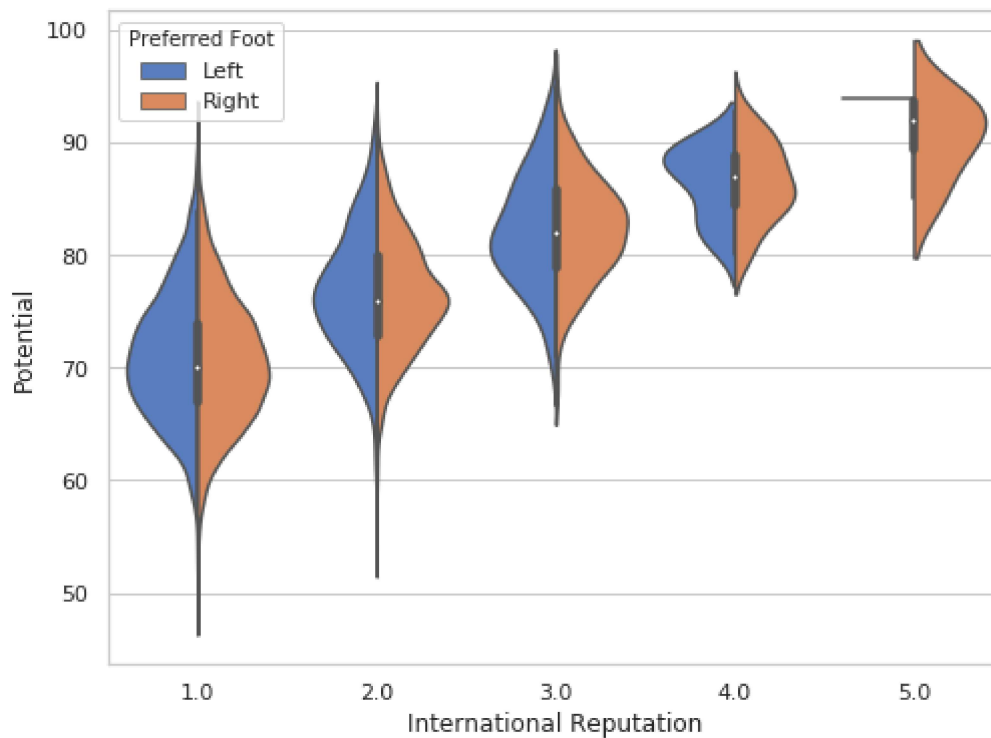
```
In [30]: f, ax = plt.subplots(figsize=(8, 6))  
sns.violinplot(x="International Reputation", y="Potential", data=fifa19)  
plt.show()
```



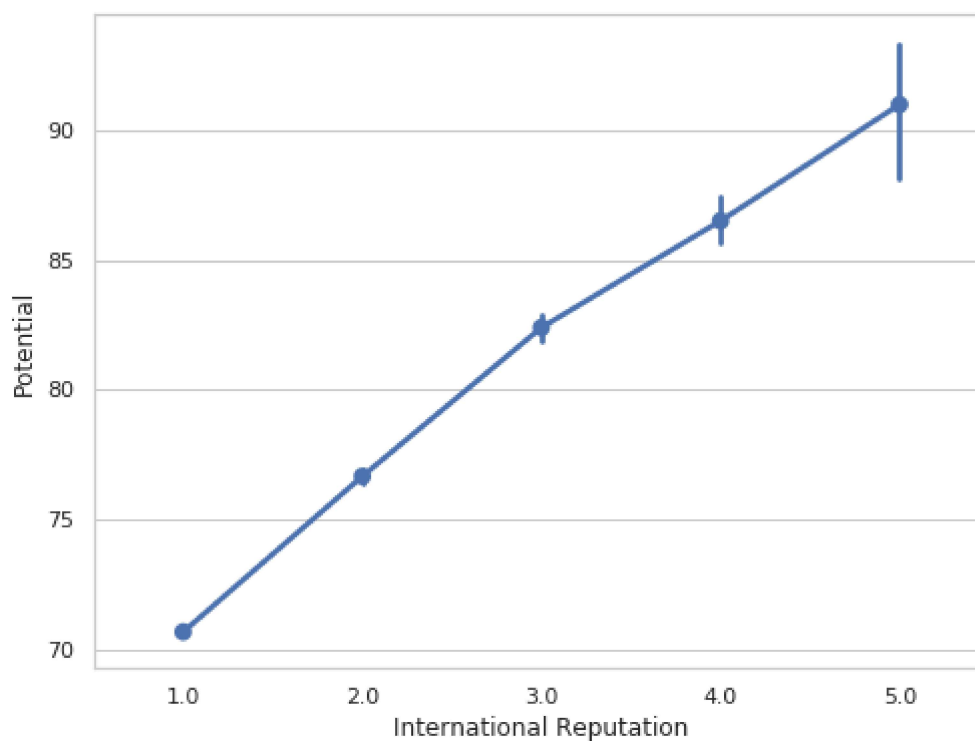
```
In [31]: f, ax = plt.subplots(figsize=(8, 6))  
sns.violinplot(x="International Reputation", y="Potential", hue="Preferred Foot", data=fifa19)  
plt.show()
```



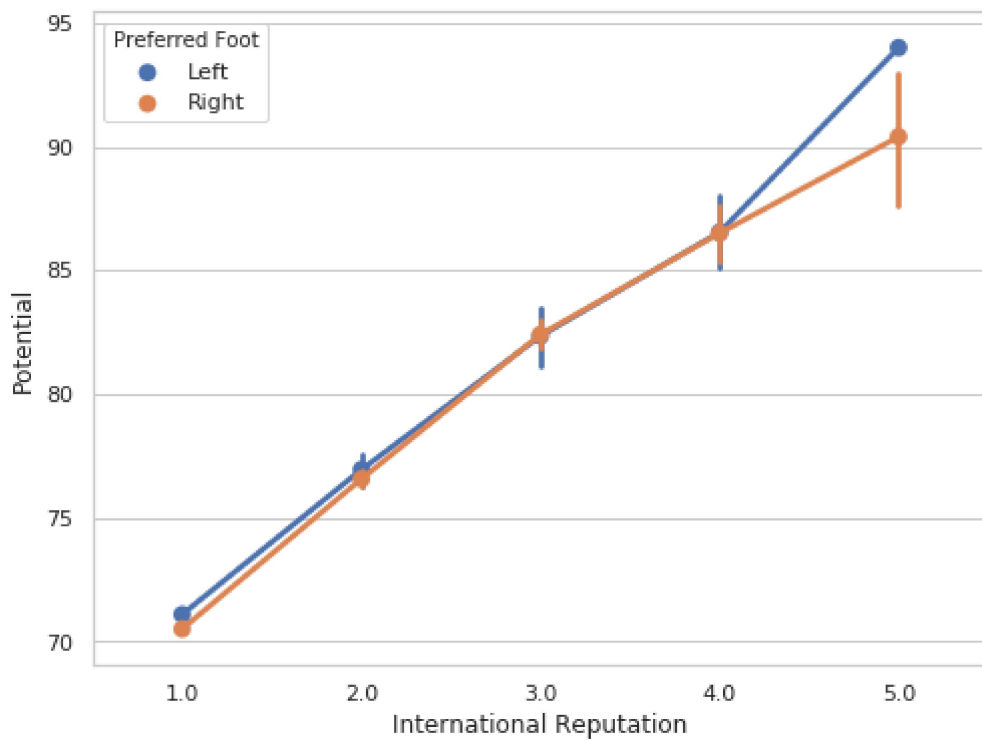
```
In [32]: f, ax = plt.subplots(figsize=(8, 6))
sns.violinplot(x="International Reputation", y="Potential", hue="Preferred Foot",
               data=fifa19, palette="muted", split=True)
plt.show()
```



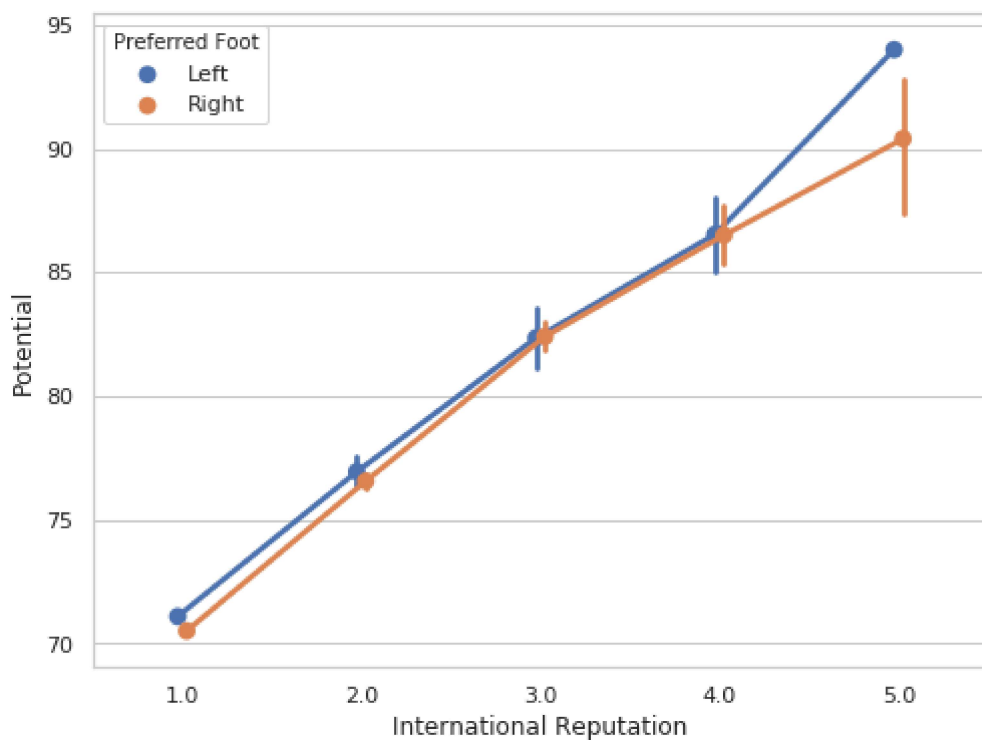
```
In [33]: f, ax = plt.subplots(figsize=(8, 6))
sns.pointplot(x="International Reputation", y="Potential", data=fifa19)
plt.show()
```



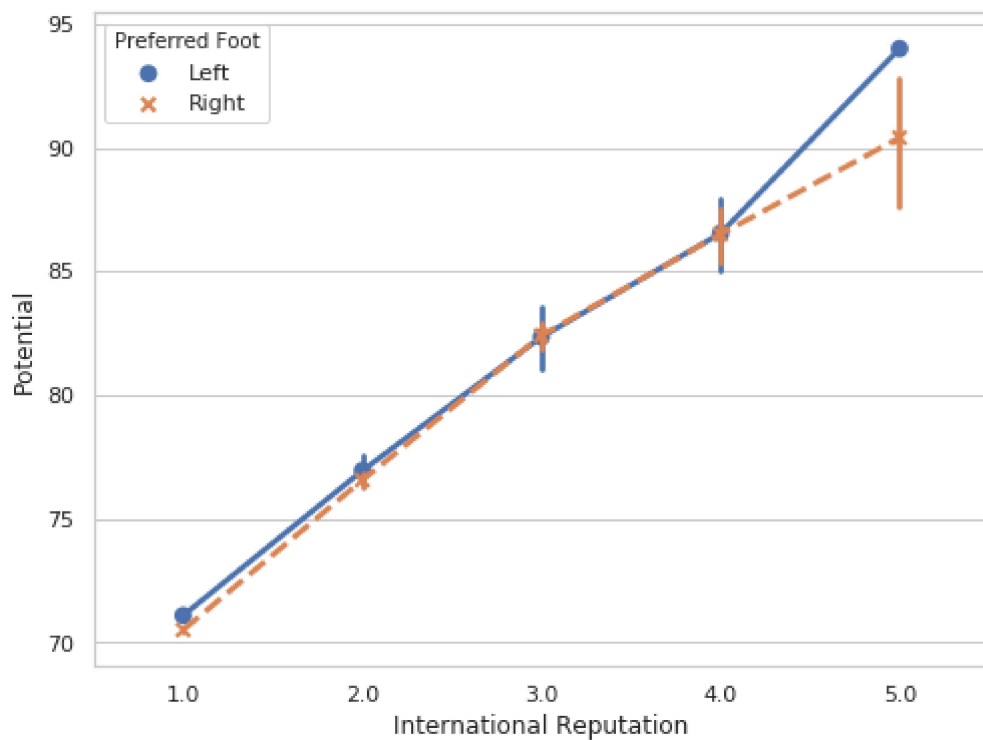
```
In [34]: f, ax = plt.subplots(figsize=(8, 6))
sns.pointplot(x="International Reputation", y="Potential", hue="Preferred Foot", data=df)
plt.show()
```



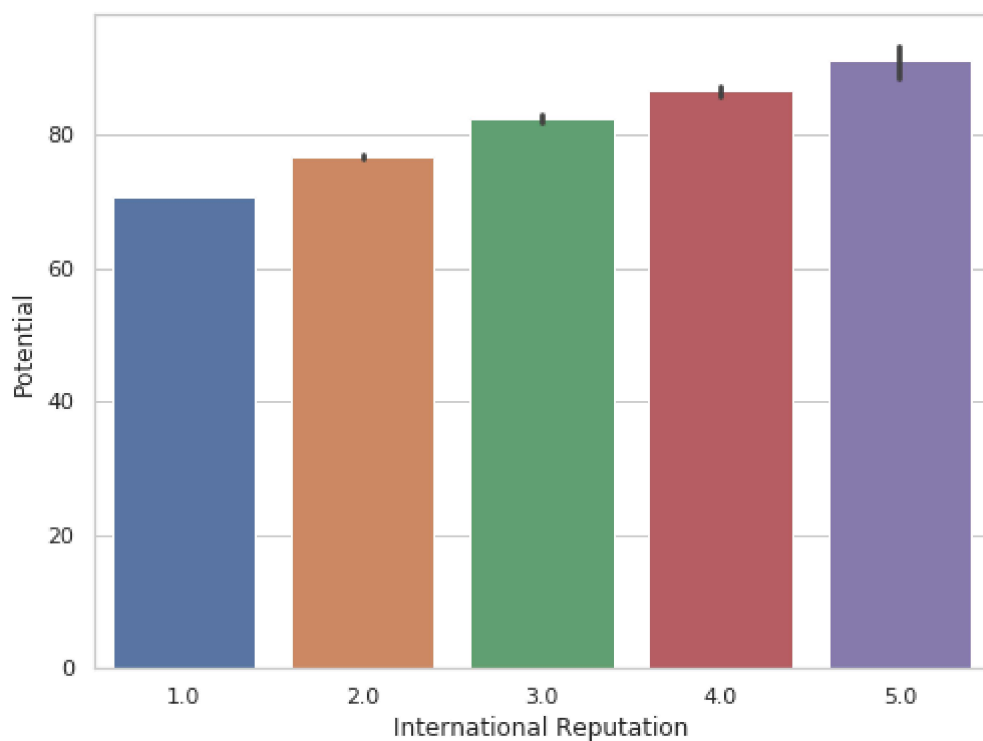
```
In [35]: f, ax = plt.subplots(figsize=(8, 6))
sns.pointplot(x="International Reputation", y="Potential", hue="Preferred Foot", data=df)
plt.show()
```



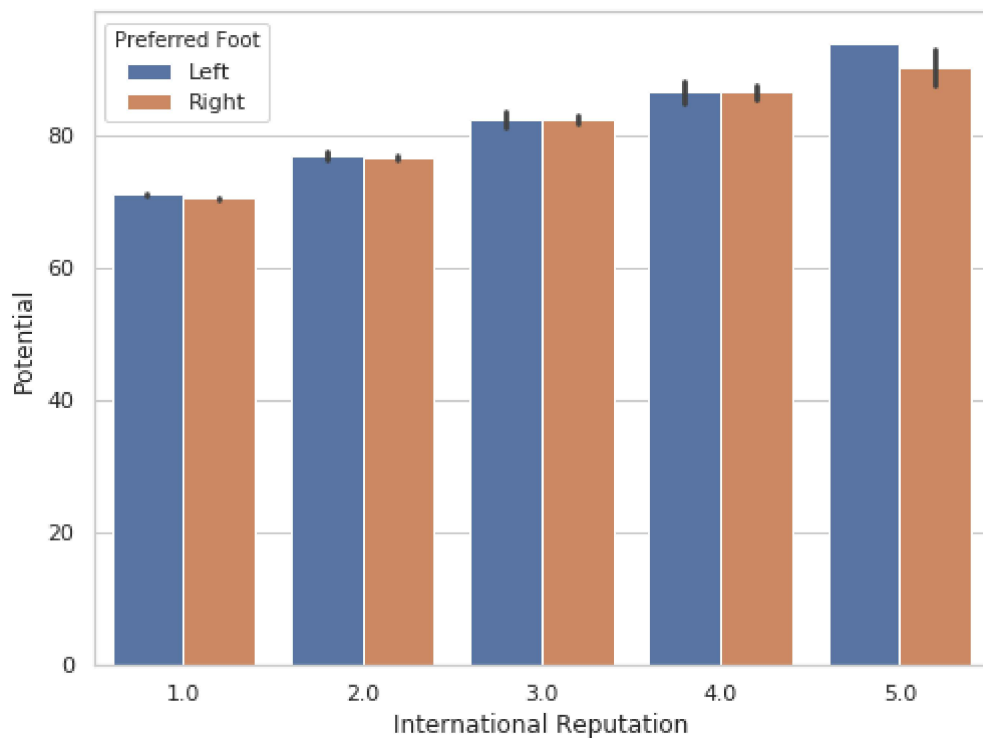
```
In [36]: f, ax = plt.subplots(figsize=(8, 6))
sns.pointplot(x="International Reputation", y="Potential", hue="Preferred Foot",
              data=fifa19, markers=["o", "x"], linestyles=["-", "--"])
plt.show()
```



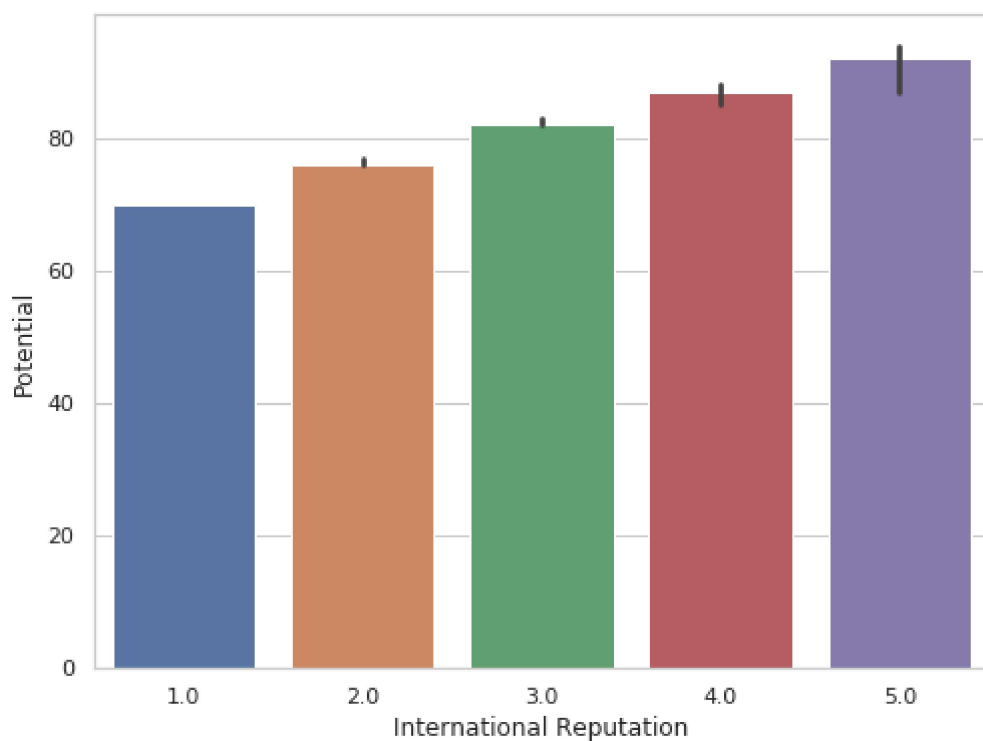
```
In [37]: f, ax = plt.subplots(figsize=(8, 6))
sns.barplot(x="International Reputation", y="Potential", data=fifa19)
plt.show()
```



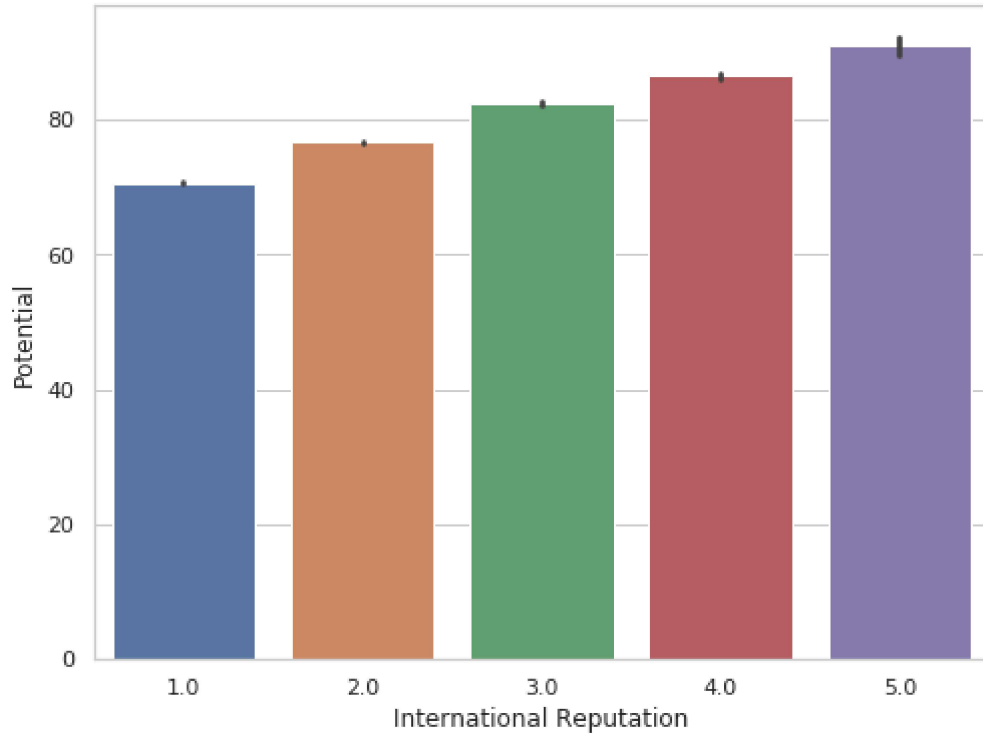
```
In [38]: f, ax = plt.subplots(figsize=(8, 6))
sns.barplot(x="International Reputation", y="Potential", hue="Preferred Foot", data=
plt.show()
```



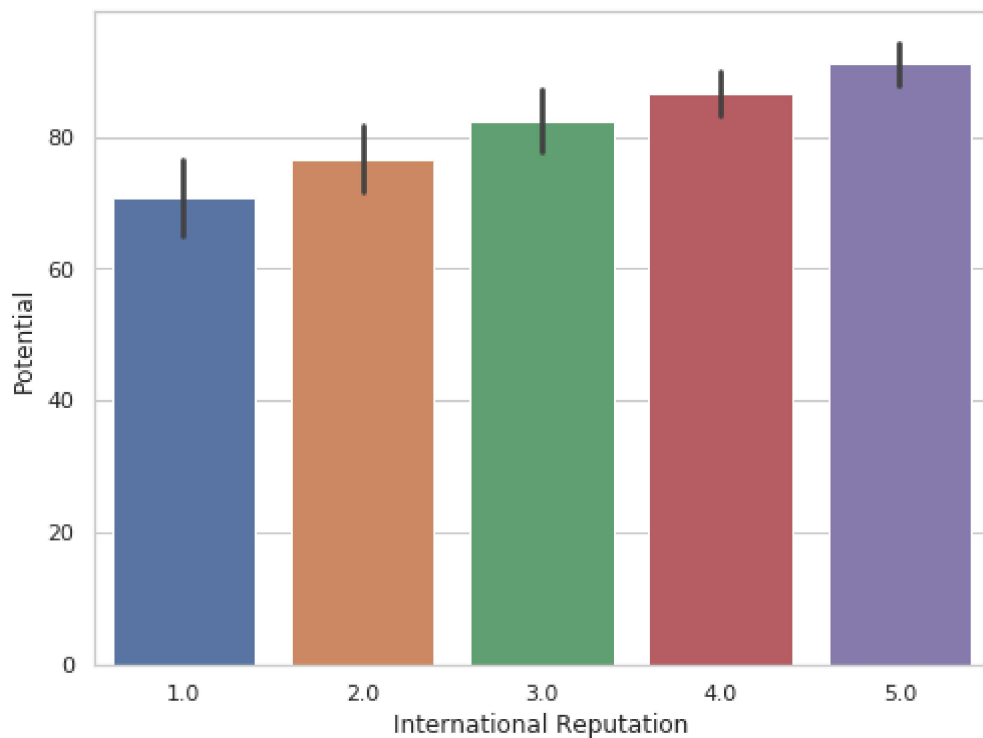
```
In [39]: from numpy import median
f, ax = plt.subplots(figsize=(8, 6))
sns.barplot(x="International Reputation", y="Potential", data=fifa19, estimator=med
plt.show()
```



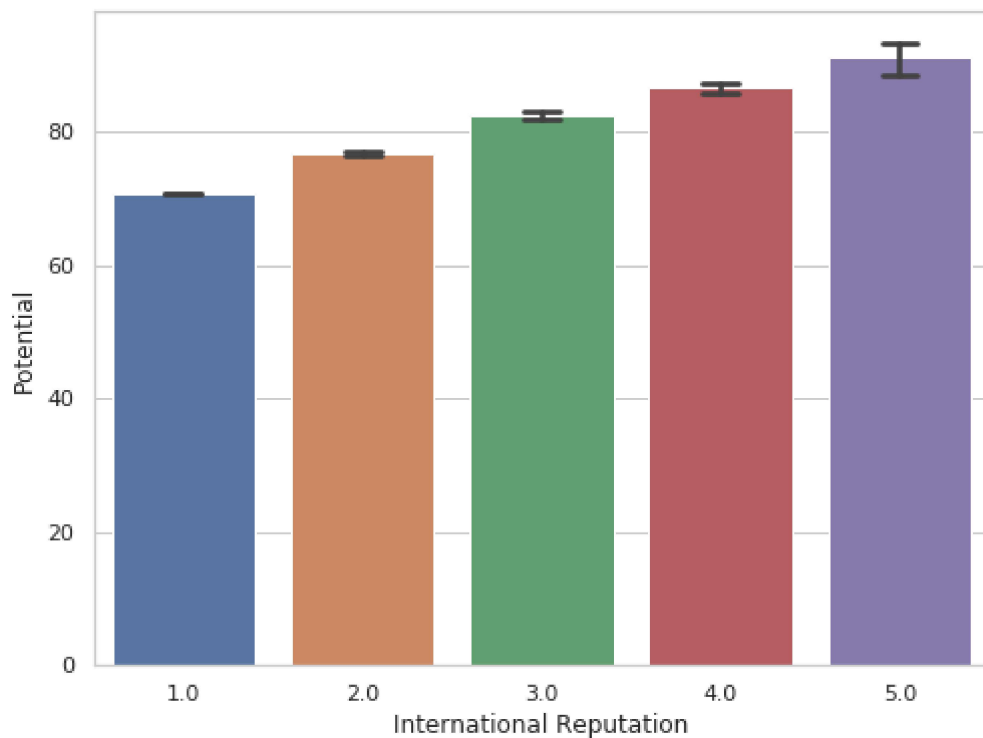
```
In [40]: f, ax = plt.subplots(figsize=(8, 6))
sns.barplot(x="International Reputation", y="Potential", data=fifa19, ci=68)
plt.show()
```



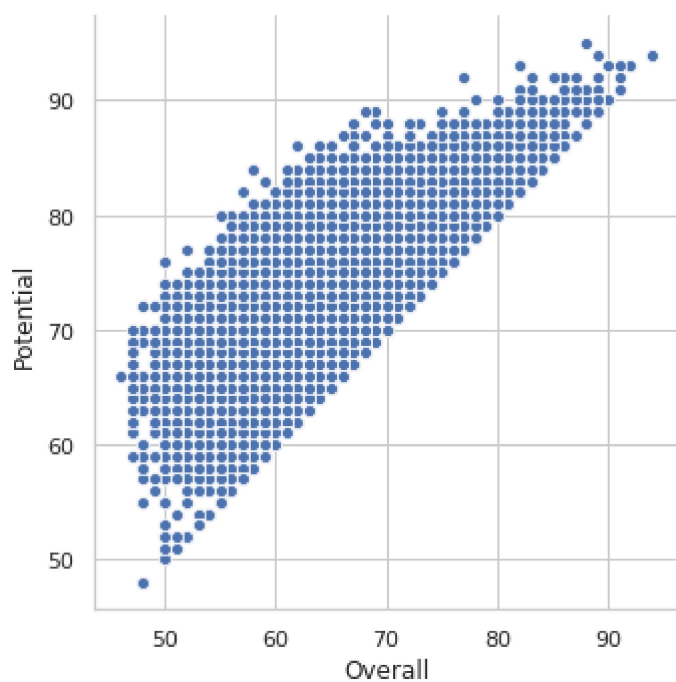
```
In [41]: f, ax = plt.subplots(figsize=(8, 6))
sns.barplot(x="International Reputation", y="Potential", data=fifa19, ci="sd")
plt.show()
```



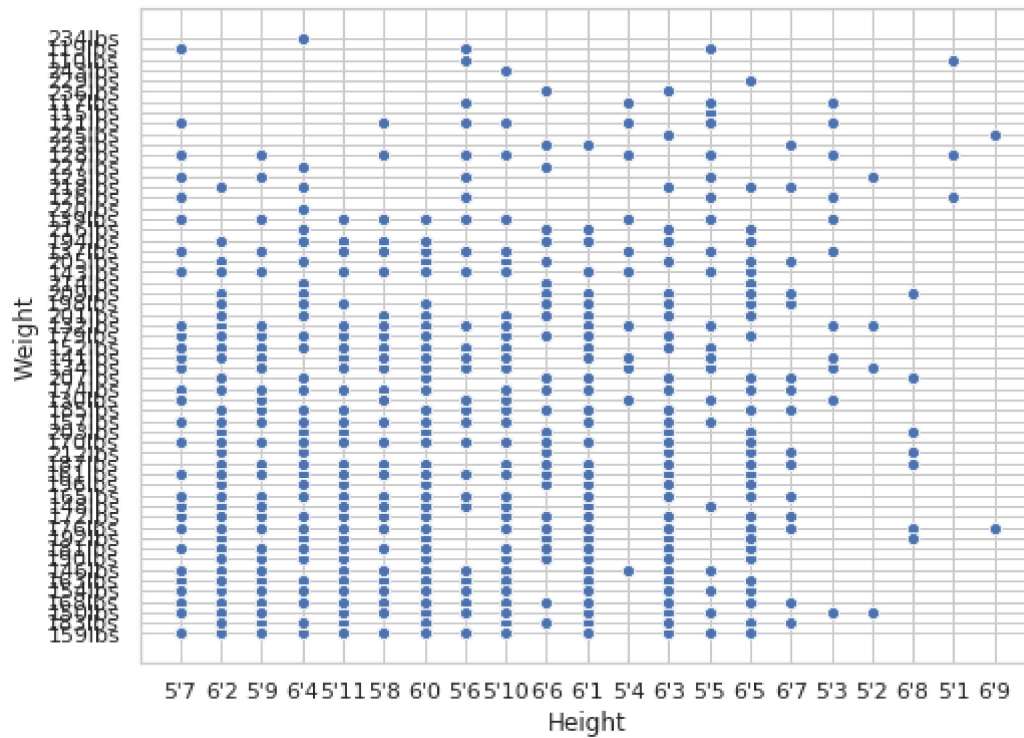
```
In [42]: f, ax = plt.subplots(figsize=(8, 6))
sns.barplot(x="International Reputation", y="Potential", data=fifa19, capsize=0.2)
plt.show()
```



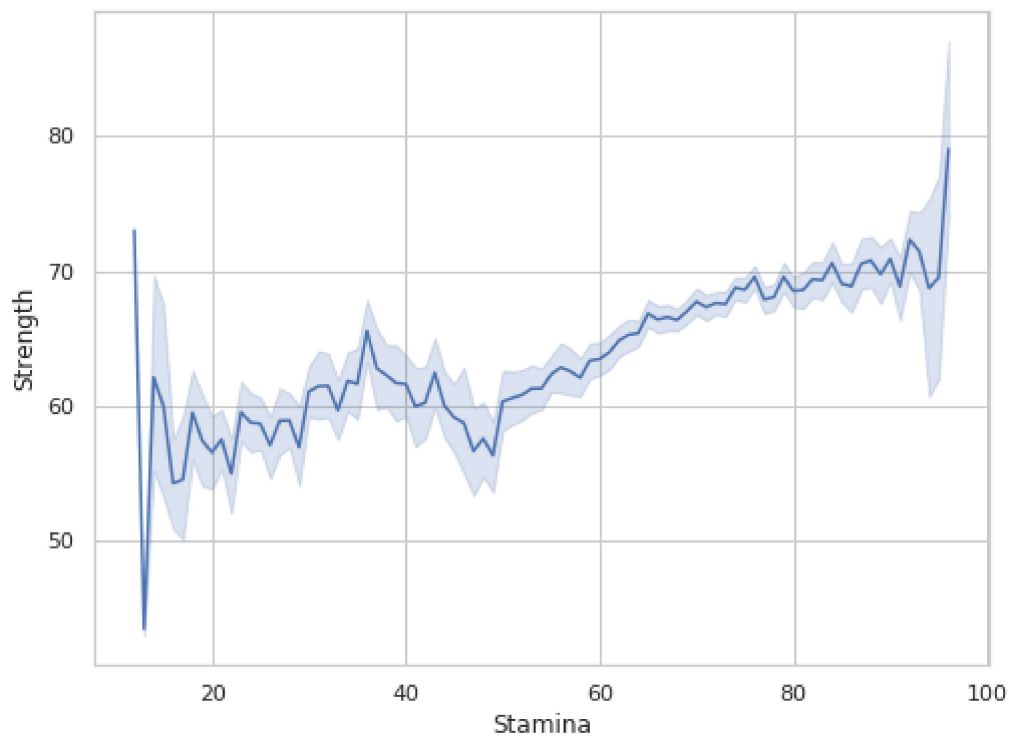
```
In [43]: g = sns.relplot(x="Overall", y="Potential", data=fifa19)
```



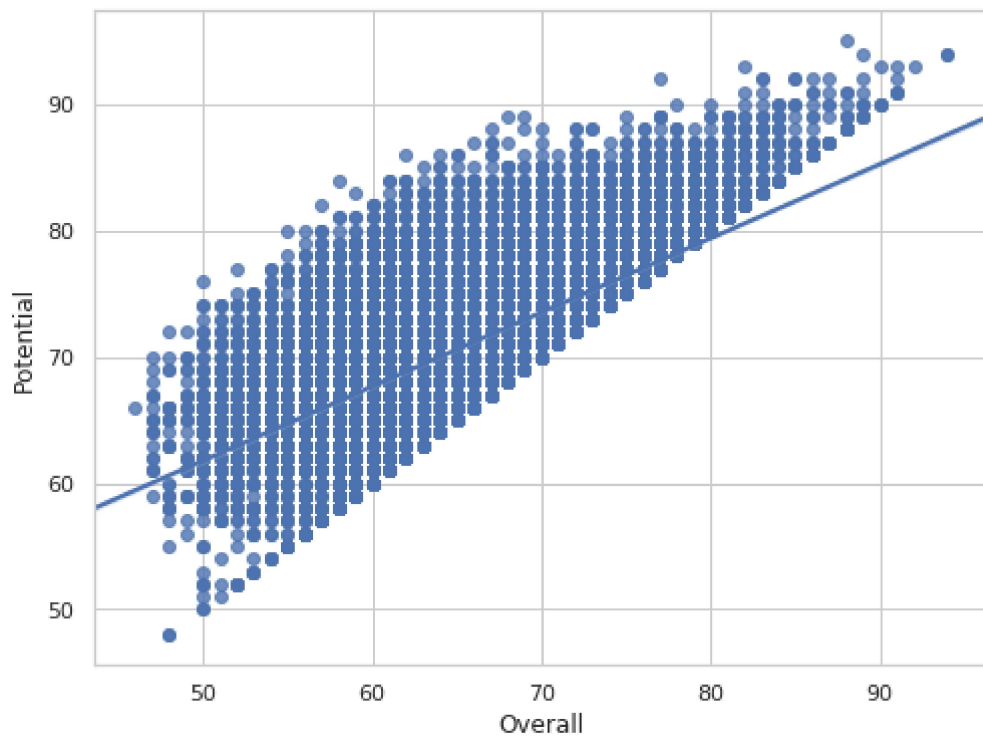
```
In [44]: f, ax = plt.subplots(figsize=(8, 6))
sns.scatterplot(x="Height", y="Weight", data=fifa19)
plt.show()
```



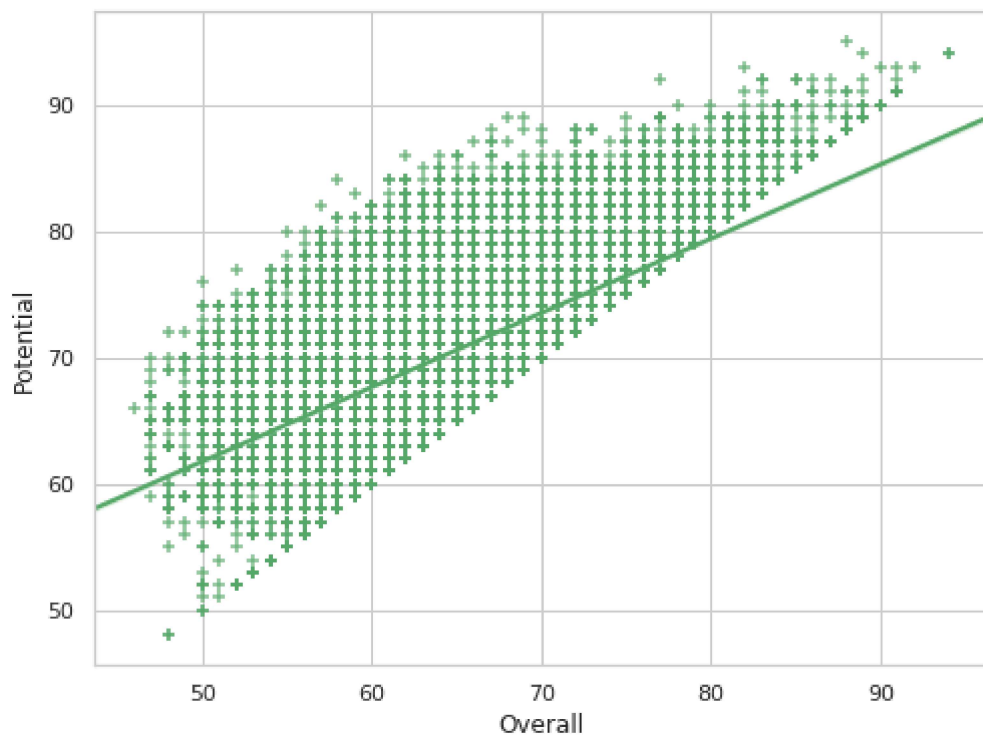
```
In [45]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.lineplot(x="Stamina", y="Strength", data=fifa19)
plt.show()
```



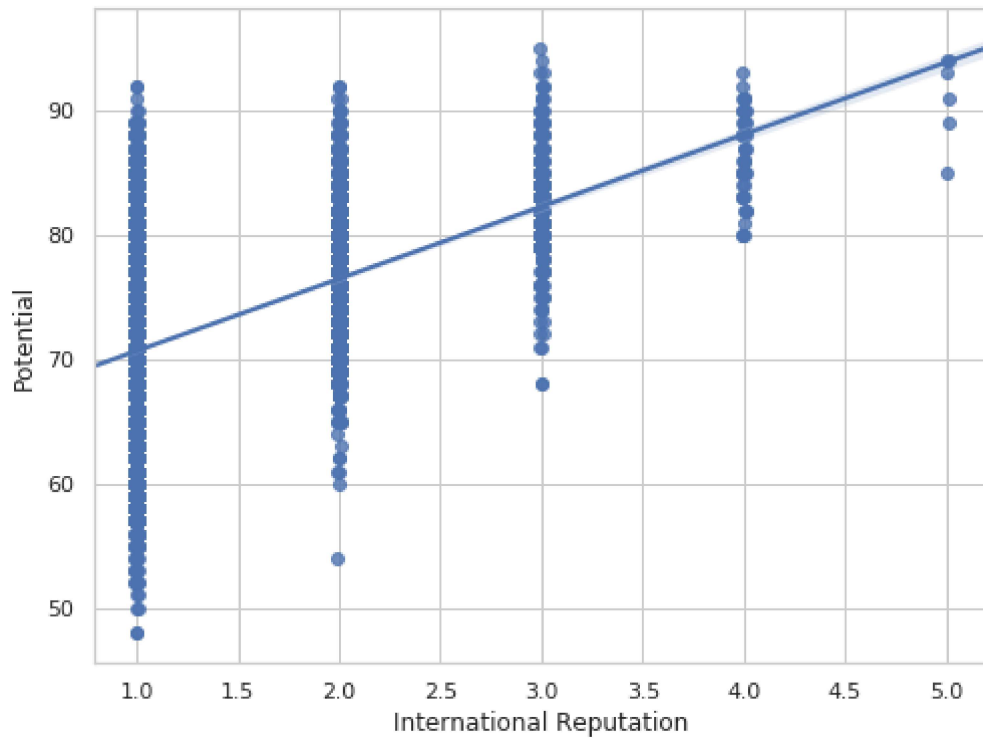
```
In [46]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.regplot(x="Overall", y="Potential", data=fifa19)
plt.show()
```



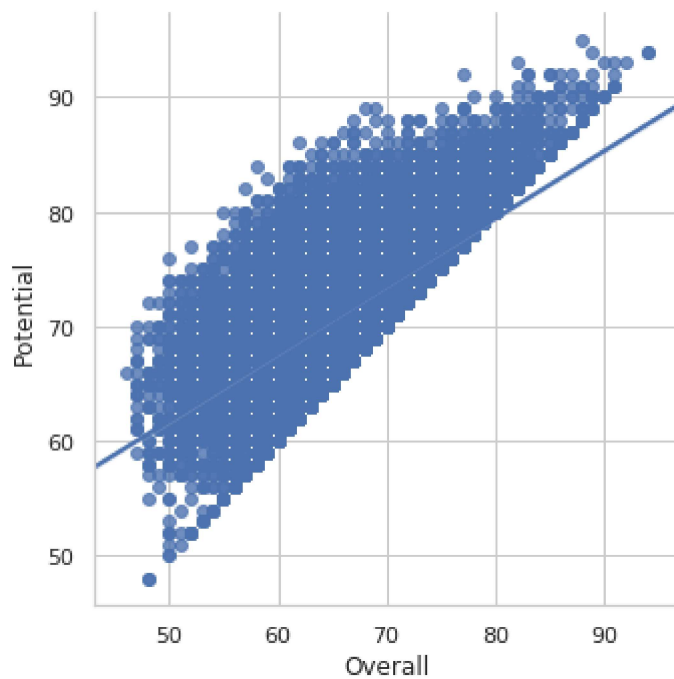
```
In [47]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.regplot(x="Overall", y="Potential", data=fifa19, color="g", marker="+")
plt.show()
```



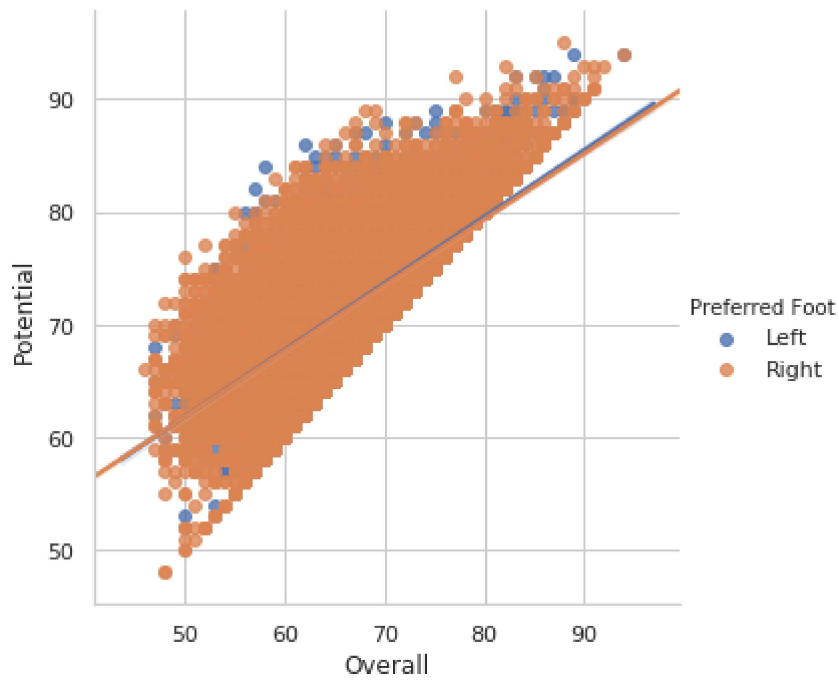
```
In [48]: f, ax = plt.subplots(figsize=(8, 6))
sns.regplot(x="International Reputation", y="Potential", data=fifa19, x_jitter=.01)
plt.show()
```



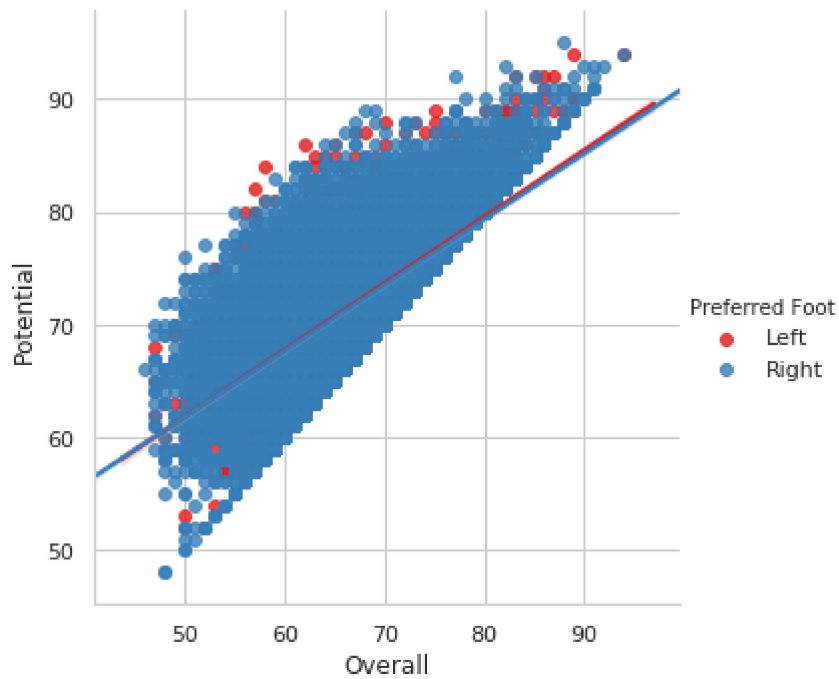
```
In [49]: g = sns.lmplot(x="Overall", y="Potential", data=fifa19)
```



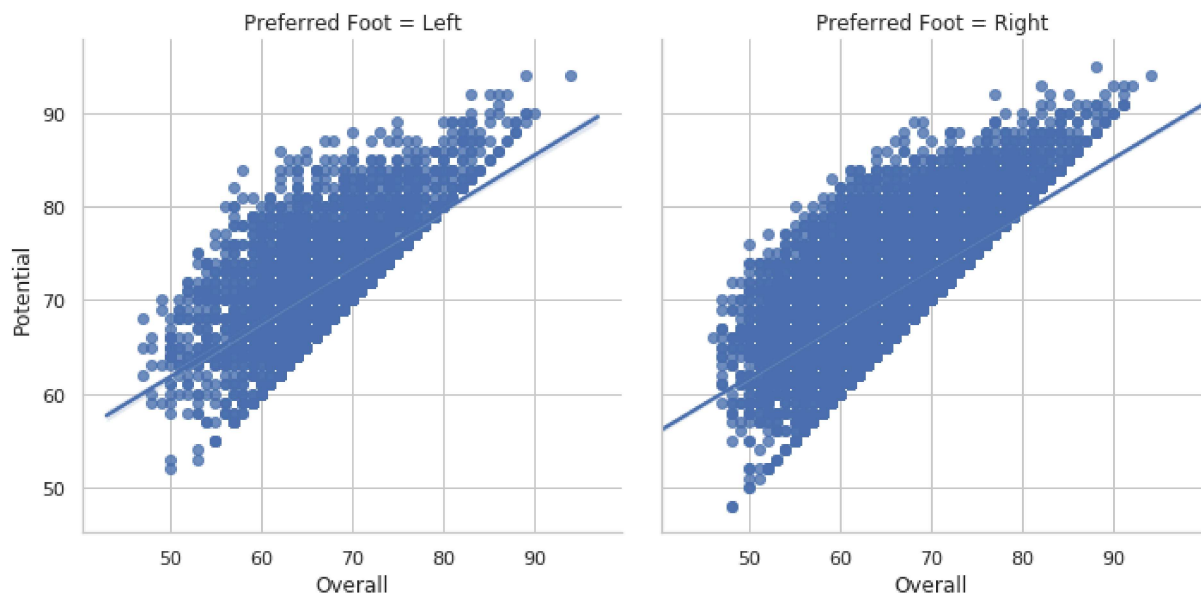
```
In [50]: g = sns.lmplot(x="Overall", y="Potential", hue="Preferred Foot", data=fifa19)
```



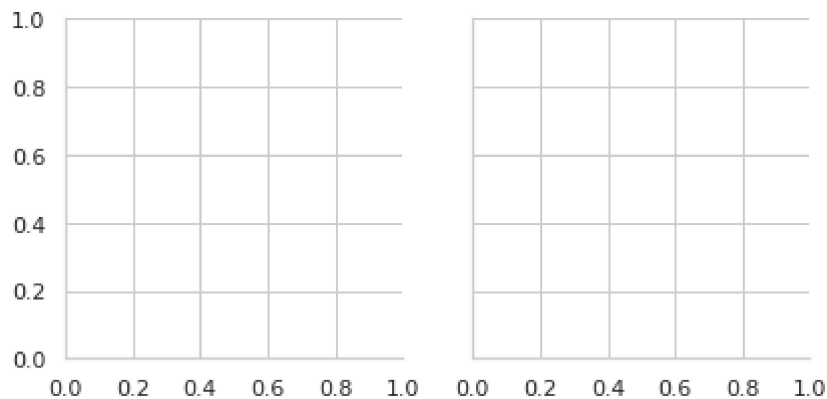
```
In [51]: g = sns.lmplot(x="Overall", y="Potential", hue="Preferred Foot", data=fifa19, palett
```



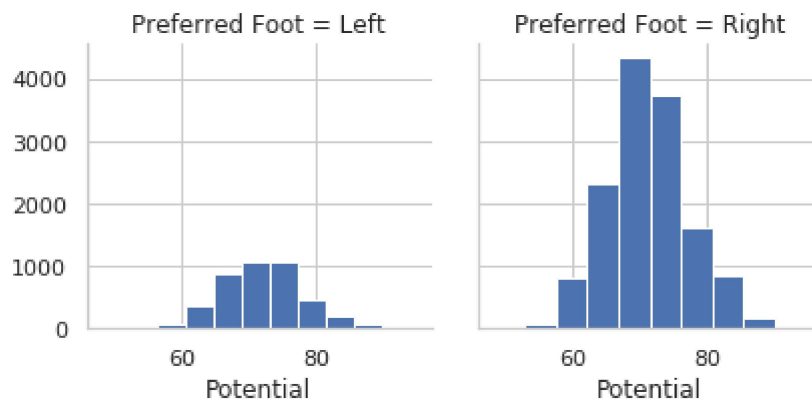
```
In [52]: g = sns.lmplot(x="Overall", y="Potential", col="Preferred Foot", data=fifa19)
```



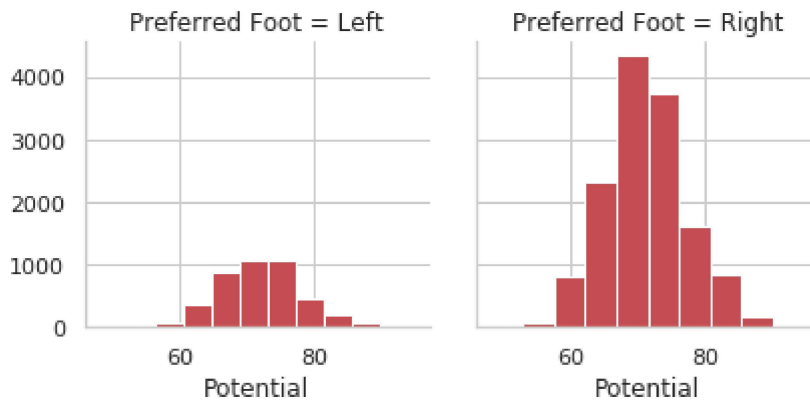
```
In [53]: g = sns.FacetGrid(fifa19, col="Preferred Foot")
```



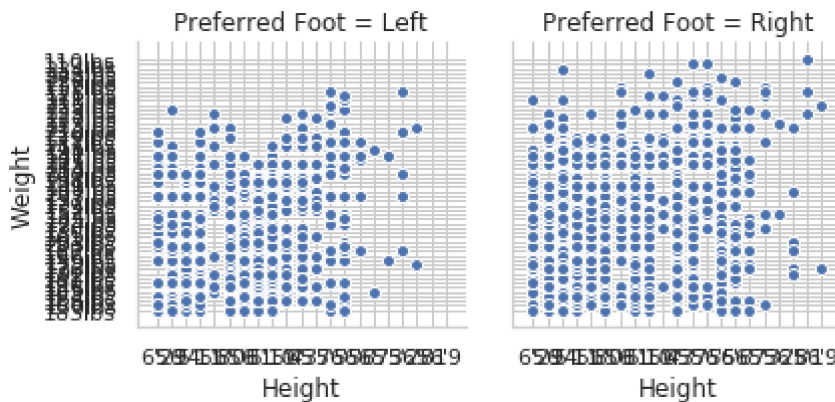
```
In [54]: g = sns.FacetGrid(fifa19, col="Preferred Foot")
g = g.map(plt.hist, "Potential")
```



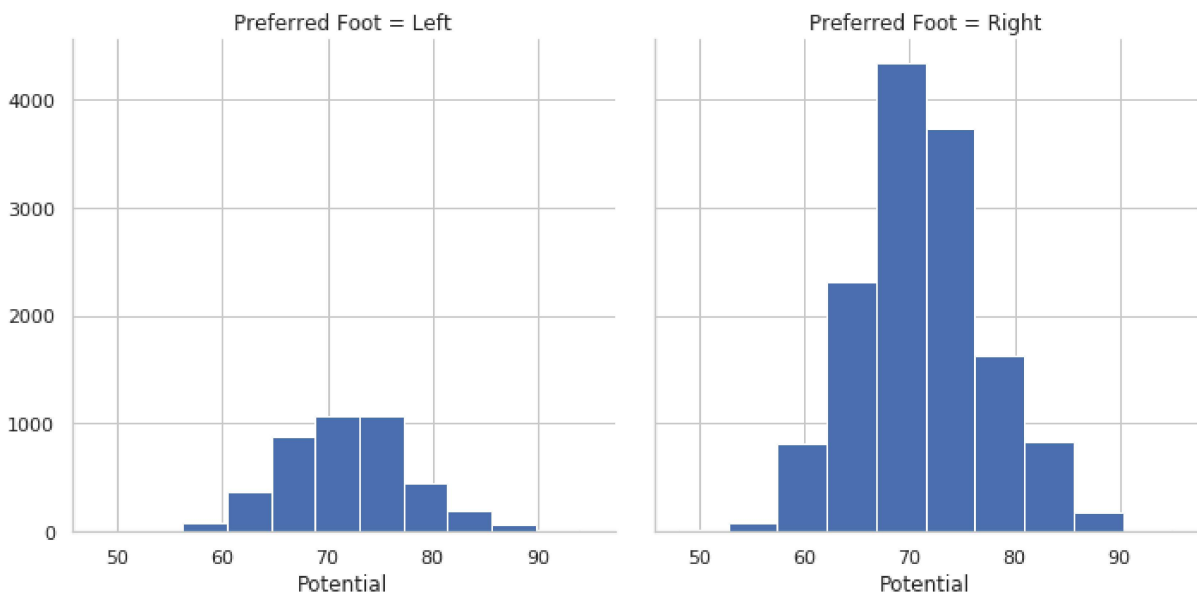
```
In [55]: g = sns.FacetGrid(fifa19, col="Preferred Foot")
g = g.map(plt.hist, "Potential", bins=10, color="r")
```



```
In [56]: g = sns.FacetGrid(fifa19, col="Preferred Foot")
g = (g.map(plt.scatter, "Height", "Weight", edgecolor="w").add_legend())
```

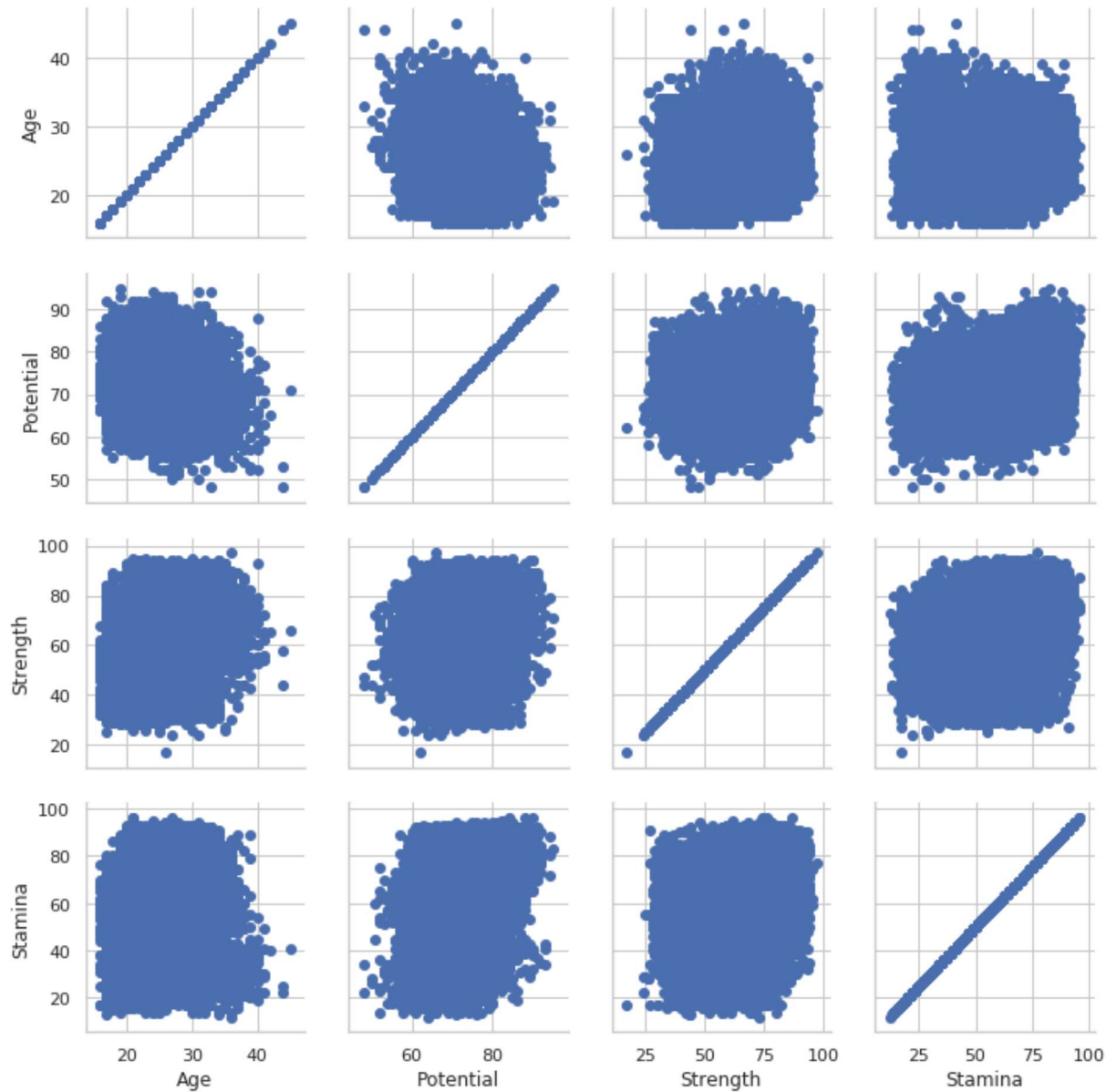


```
In [57]: g = sns.FacetGrid(fifa19, col="Preferred Foot", height=5, aspect=1)
g = g.map(plt.hist, "Potential")
```

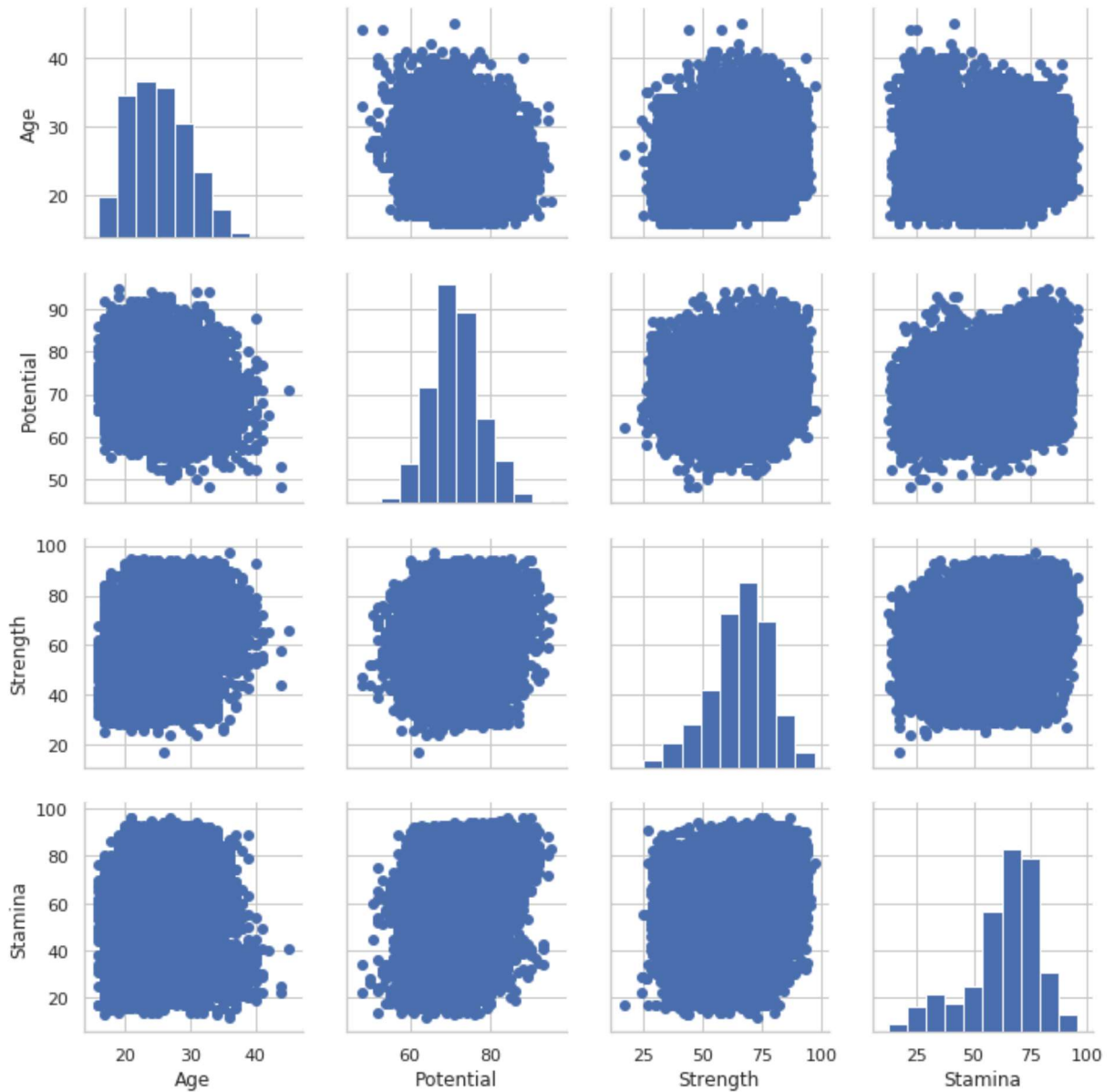


```
In [58]: fifa19_new = fifa19[['Age', 'Potential', 'Strength', 'Stamina', 'Preferred Foot']]
```

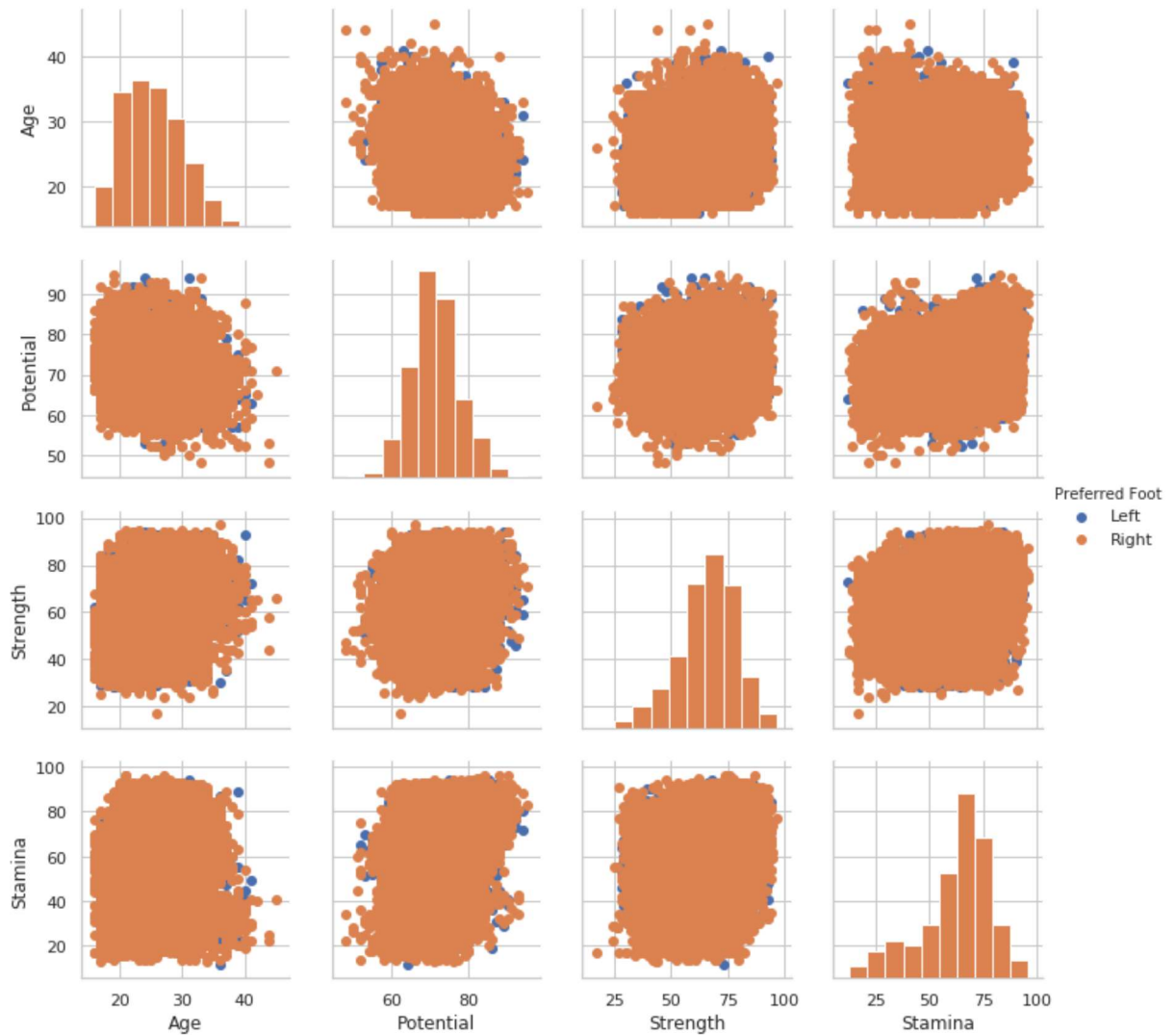
```
In [59]: g = sns.PairGrid(fifa19_new)
g = g.map(plt.scatter)
```



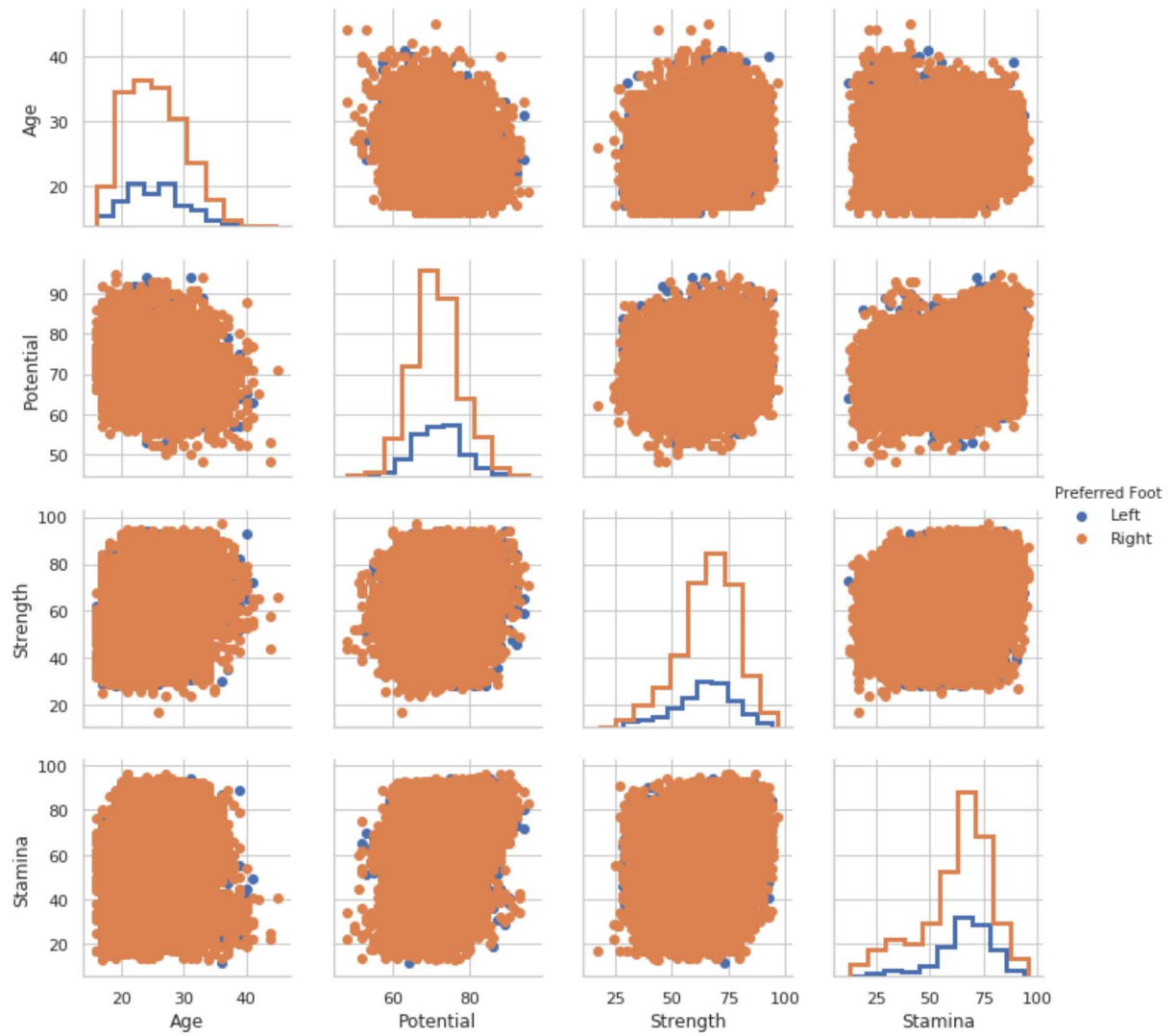
```
In [60]: g = sns.PairGrid(fifa19_new)
g = g.map_diag(plt.hist)
g = g.map_offdiag(plt.scatter)
```



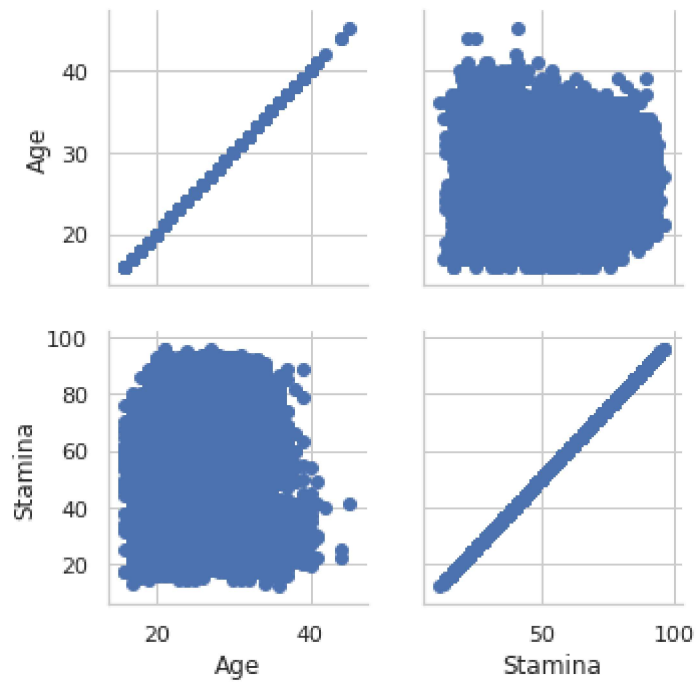
```
In [61]: g = sns.PairGrid(fifa19_new, hue="Preferred Foot")
g = g.map_diag(plt.hist)
g = g.map_offdiag(plt.scatter)
g = g.add_legend()
```



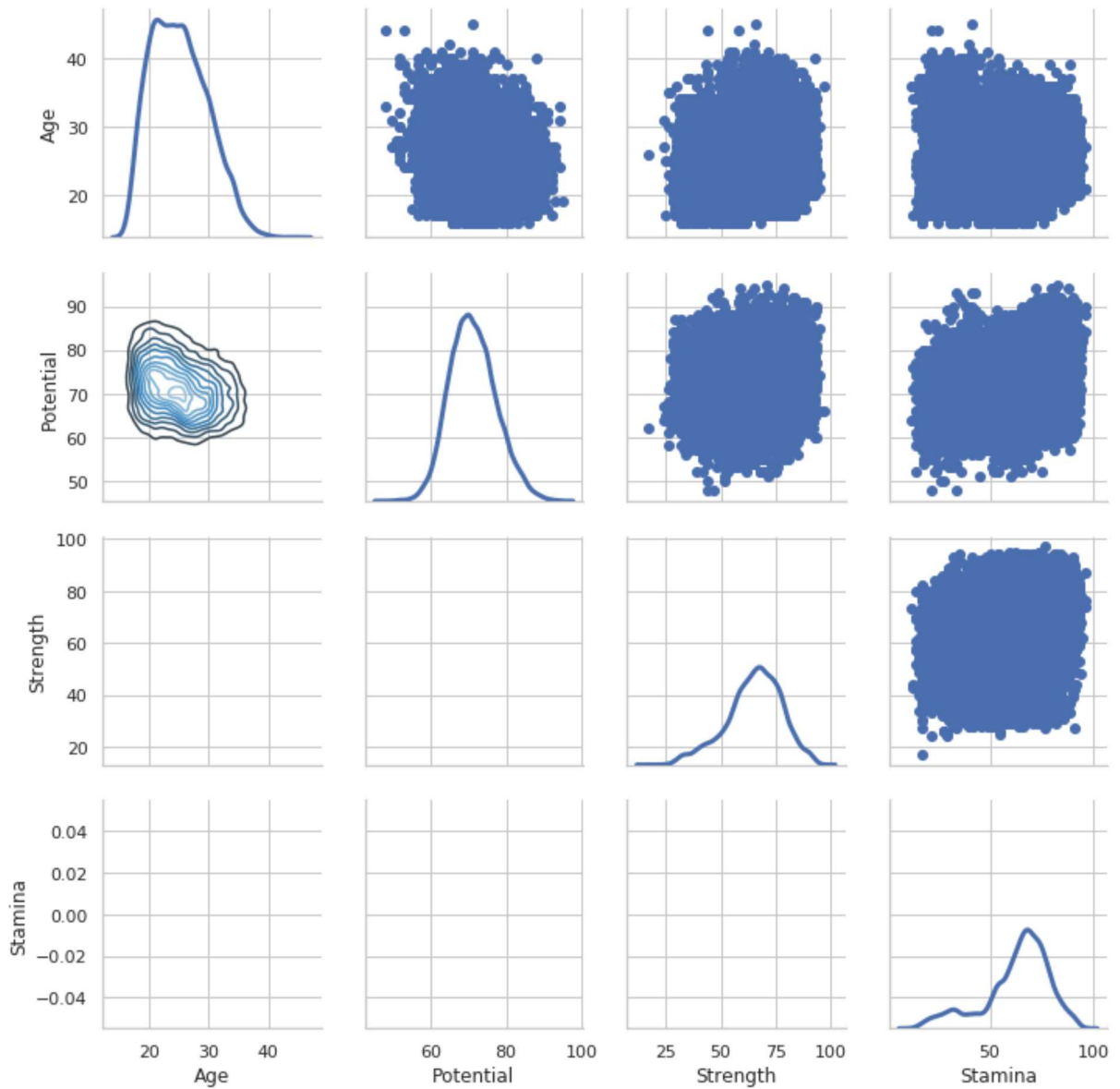
```
In [62]: g = sns.PairGrid(fifa19_new, hue="Preferred Foot")
g = g.map_diag(plt.hist, histtype="step", linewidth=3)
g = g.map_offdiag(plt.scatter)
g = g.add_legend()
```



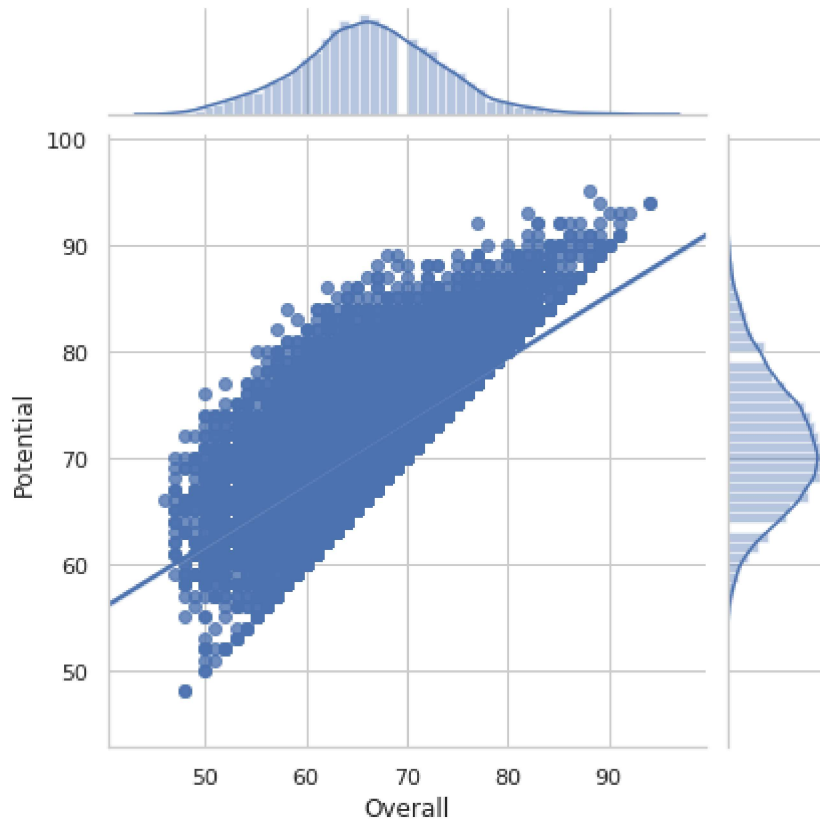
```
In [63]: g = sns.PairGrid(fifa19_new, vars=['Age', 'Stamina'])
g = g.map(plt.scatter)
```



```
In [64]: g = sns.PairGrid(fifa19_new)
g = g.map_upper(plt.scatter)
g = g.map_lower(sns.kdeplot, cmap="Blues_d")
g = g.map_diag(sns.kdeplot, lw=3, legend=False)
```

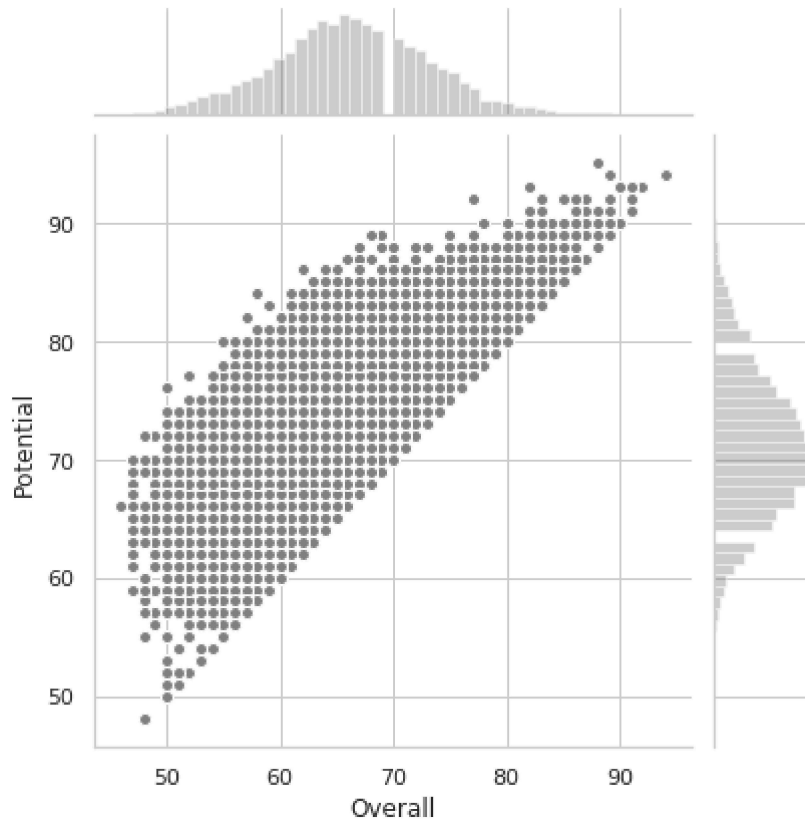


```
In [65]: g = sns.JointGrid(x="Overall", y="Potential", data=fifa19)
g = g.plot(sns.regplot, sns.distplot)
```

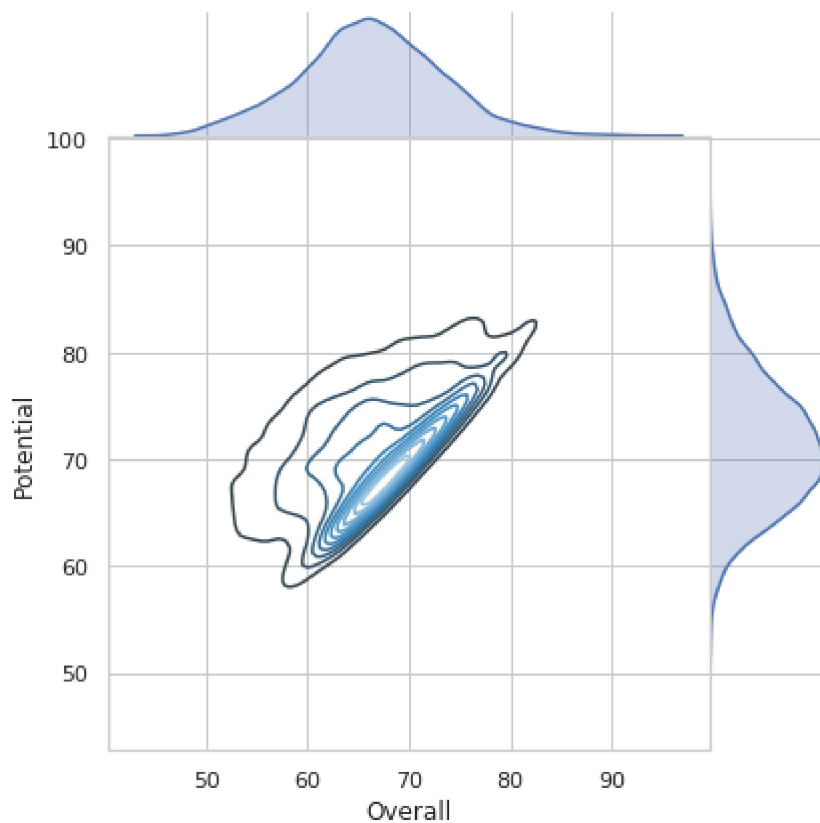


```
In [66]: import matplotlib.pyplot as plt
```

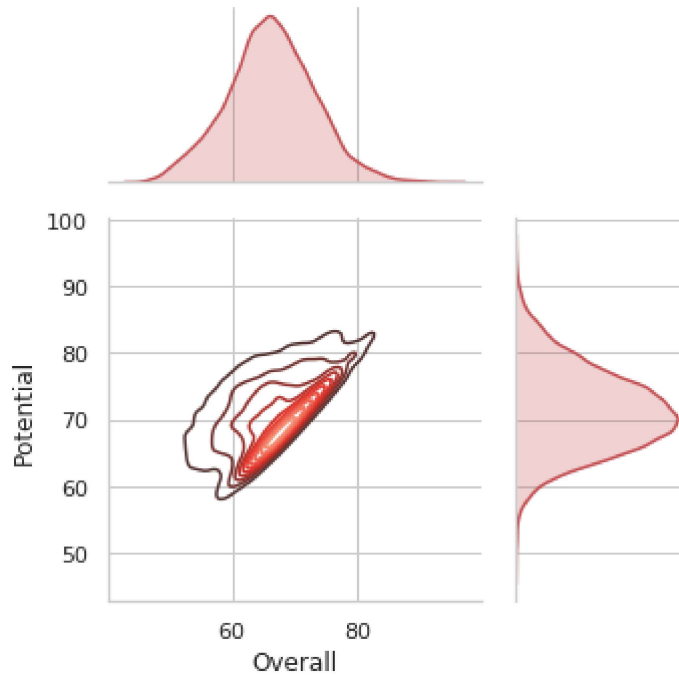
```
In [67]: g = sns.JointGrid(x="Overall", y="Potential", data=fifa19)
g = g.plot_joint(plt.scatter, color=".5", edgecolor="white")
g = g.plot_marginals(sns.distplot, kde=False, color=".5")
```



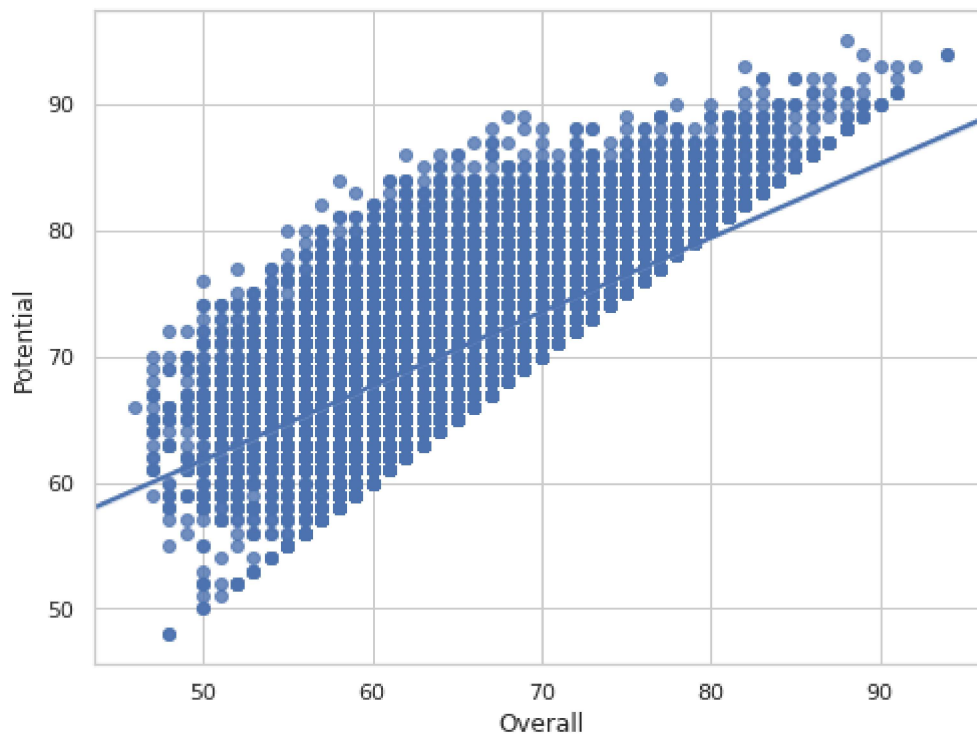
```
In [68]: g = sns.JointGrid(x="Overall", y="Potential", data=fifa19, space=0)
g = g.plot_joint(sns.kdeplot, cmap="Blues_d")
g = g.plot_marginals(sns.kdeplot, shade=True)
```



```
In [69]: g = sns.JointGrid(x="Overall", y="Potential", data=fifa19, height=5, ratio=2)
g = g.plot_joint(sns.kdeplot, cmap="Reds_d")
g = g.plot_marginals(sns.kdeplot, color="r", shade=True)
```

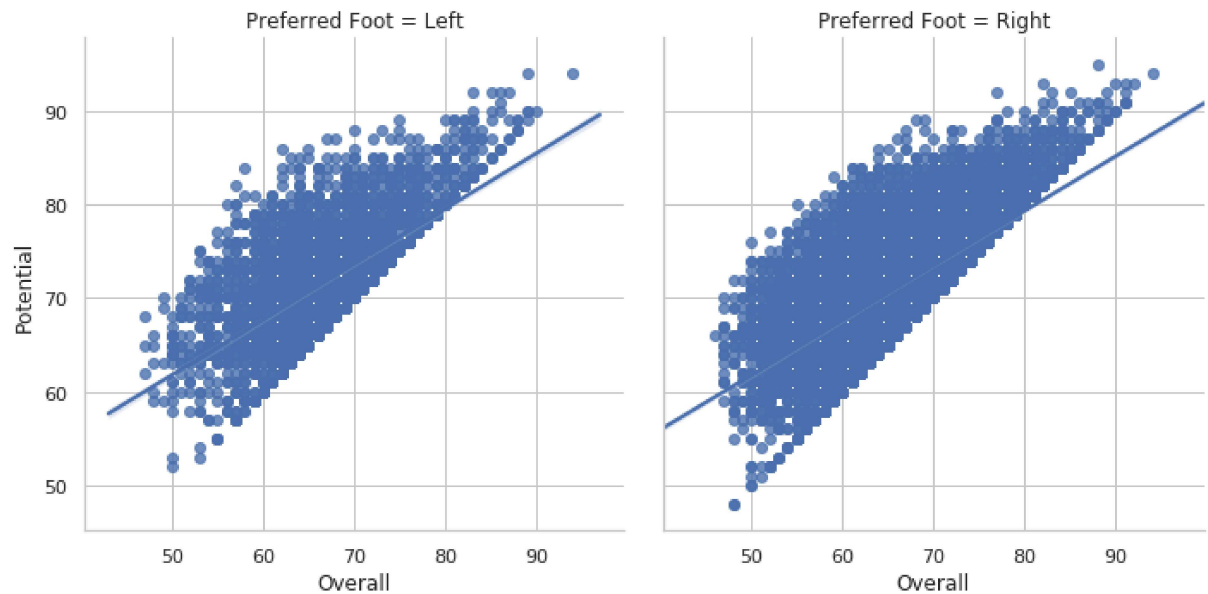


```
In [70]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.regplot(x="Overall", y="Potential", data=fifa19);
```



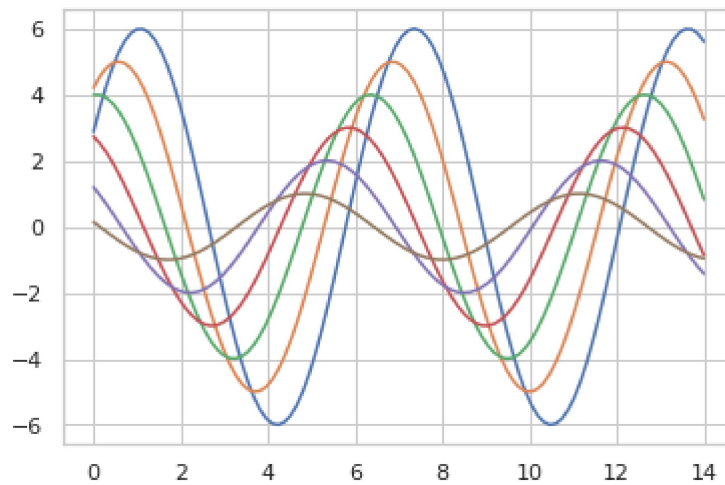
```
In [71]: sns.lmplot(x="Overall", y="Potential", col="Preferred Foot", data=fifa19, col_wrap=
```

```
Out[71]: <seaborn.axisgrid.FacetGrid at 0x7fc7a26a51d0>
```

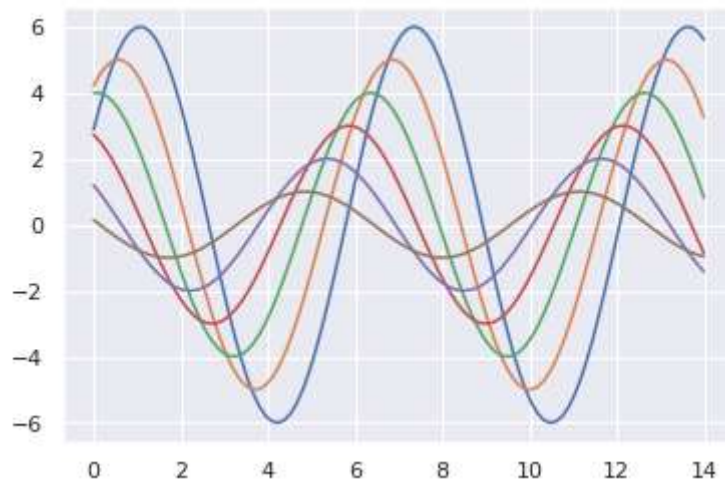


```
In [72]: def sinplot(flip=1):
          x = np.linspace(0, 14, 100)
          for i in range(1, 7):
              plt.plot(x, np.sin(x + i * .5) * (7 - i) * flip)
```

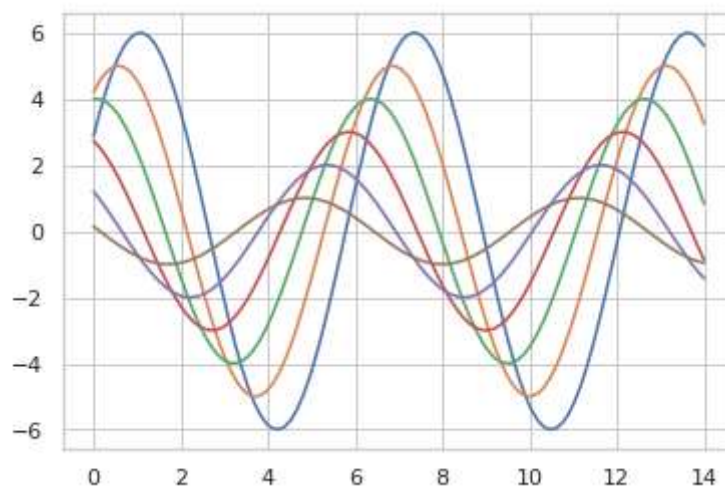
```
In [73]: sinplot()
```



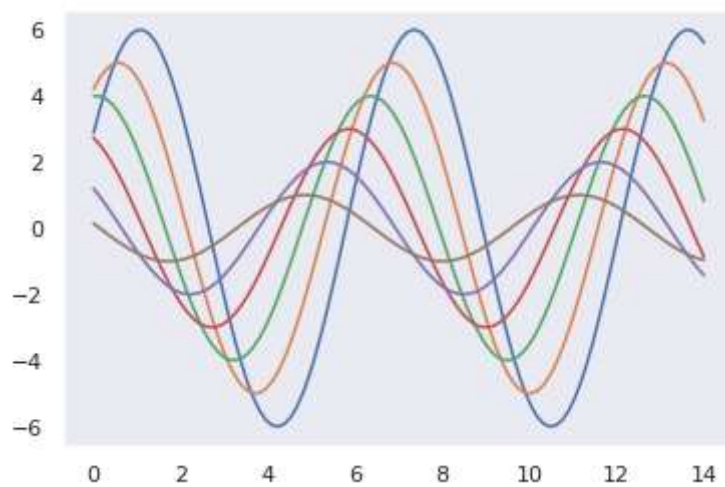
```
In [74]: sns.set()
          sinplot()
```



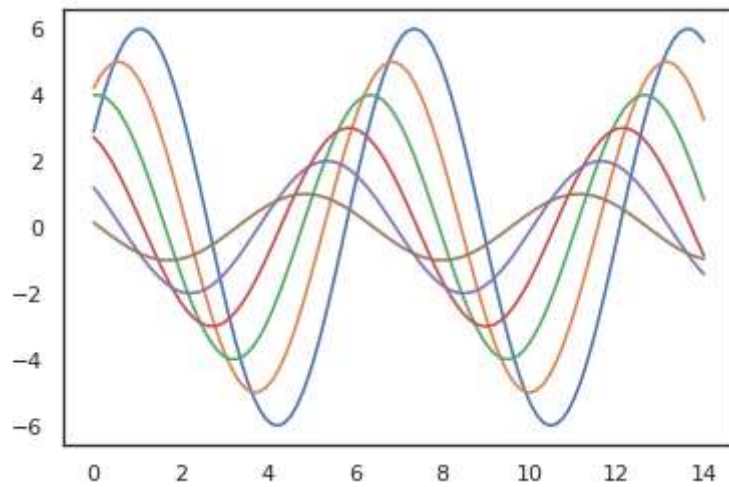
```
In [75]: sns.set_style("whitegrid")  
sinplot()
```



```
In [76]: sns.set_style("dark")  
sinplot()
```



```
In [77]: sns.set_style("white")  
sinplot()
```



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In [ ]:
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In [ ]:
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In [ ]:
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In [ ]:
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